UltraEdge Hyper-V Installation Guide

Reference : RA 477

Description

This documentation provides instructions for creating an UltraEdge simulator using Microsoft Hyper-V.

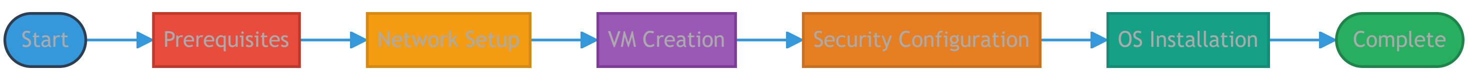
- Installation Overview - Visual guide to the complete process
- Overview
- Hyper-V activation
- Nested VM
- Installation

UltraEdge Hyper-V - Installation Overview

This page provides a visual overview of the UltraEdge simulator installation process using Microsoft Hyper-V.

High-Level Installation Process

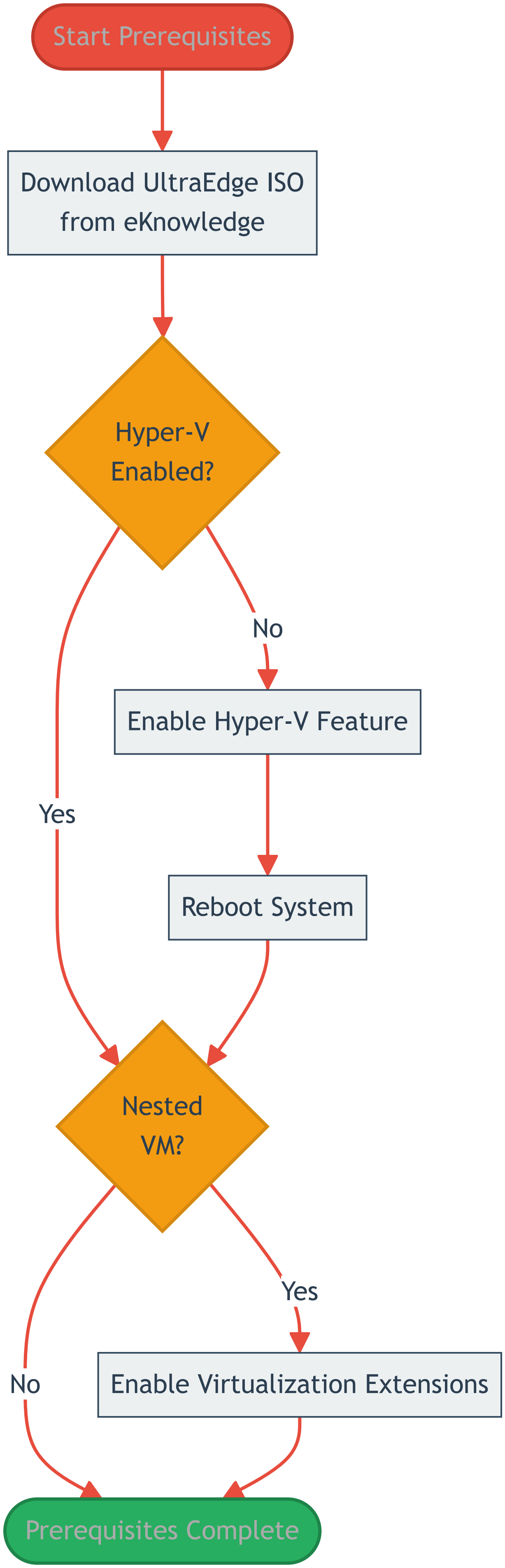
The following diagram shows the main phases of the installation. Click on any phase to jump to detailed steps.



> **TIP:** Click on any phase in the diagram above to jump to detailed steps for that phase.

Prerequisites Detail

This phase ensures your system has everything needed to begin the installation.


Start Prerequisites
Download UltraEdge ISO from eKnowledge
Hyper-V Enabled?
No
Yes
Enable Hyper-V Feature
Reboot System
Nested VM?
Yes
No
Enable Virtualization Extensions
Prerequisites Complete

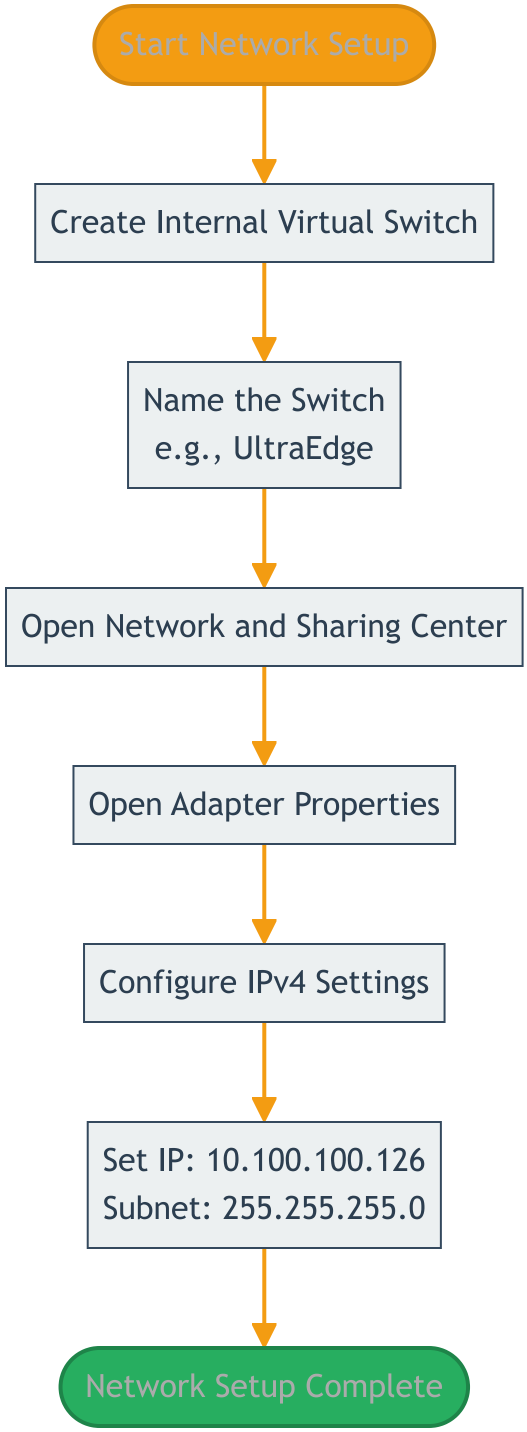
Key Steps:

- Download the UltraEdge ISO file from eKnowledge (approximately 2GB uncompressed)
- Enable Hyper-V feature in Windows if not already enabled
- Reboot system after enabling Hyper-V
- If using nested VMs, enable virtualization extensions on the host VM

↑ Back to overview

Network Setup Detail

This phase configures the virtual network for the UltraEdge VM.



Key Steps:

- Create a new Internal Virtual Switch in Hyper-V Manager
- Name it appropriately (e.g., "UltraEdge")
- Configure the network adapter with static IP: 10.100.100.126/24

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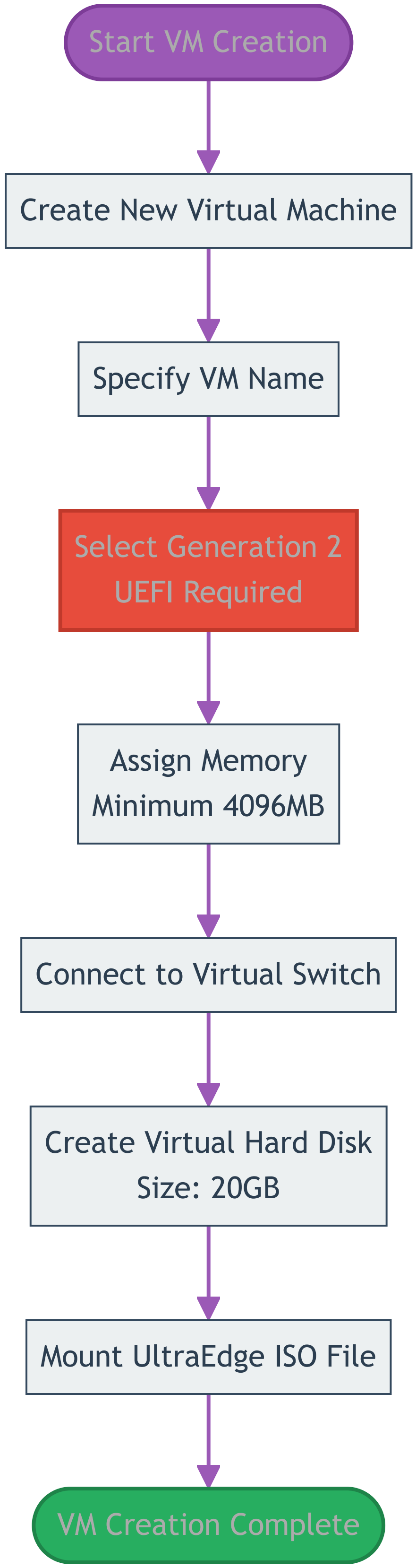
VM Creation Detail

This phase creates and configures the virtual machine using the Hyper-V wizard.



Key Steps:

- Launch the New Virtual Machine wizard in Hyper-V Manager

- Name your virtual machine
- **Important:** Select Generation 2 (required for UEFI boot)
- Allocate at least 4096MB of memory
- Connect to the previously created virtual switch
- Create a 20GB virtual hard disk
- Attach the UltraEdge ISO file

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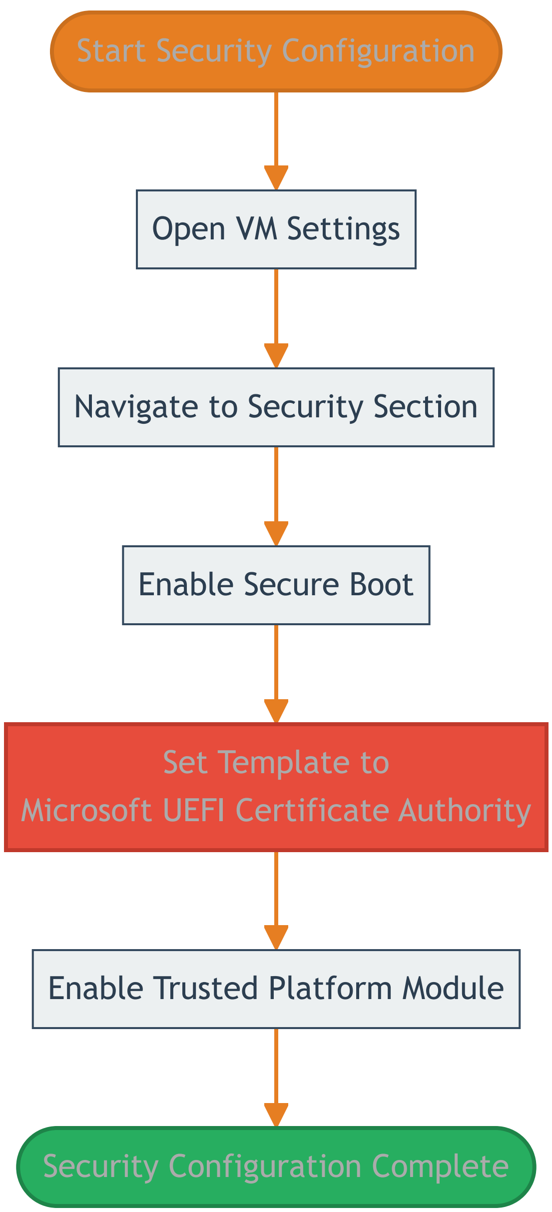
Security Configuration Detail

This phase configures essential security settings before the first boot.



Key Steps:

- Open the VM settings
- Navigate to Security under Hardware
- Ensure Secure Boot is enabled
- **Critical:** Change template from "Microsoft Windows" to "Microsoft UEFI Certificate Authority" for Linux
- Enable Trusted Platform Module (TPM) for secure connections

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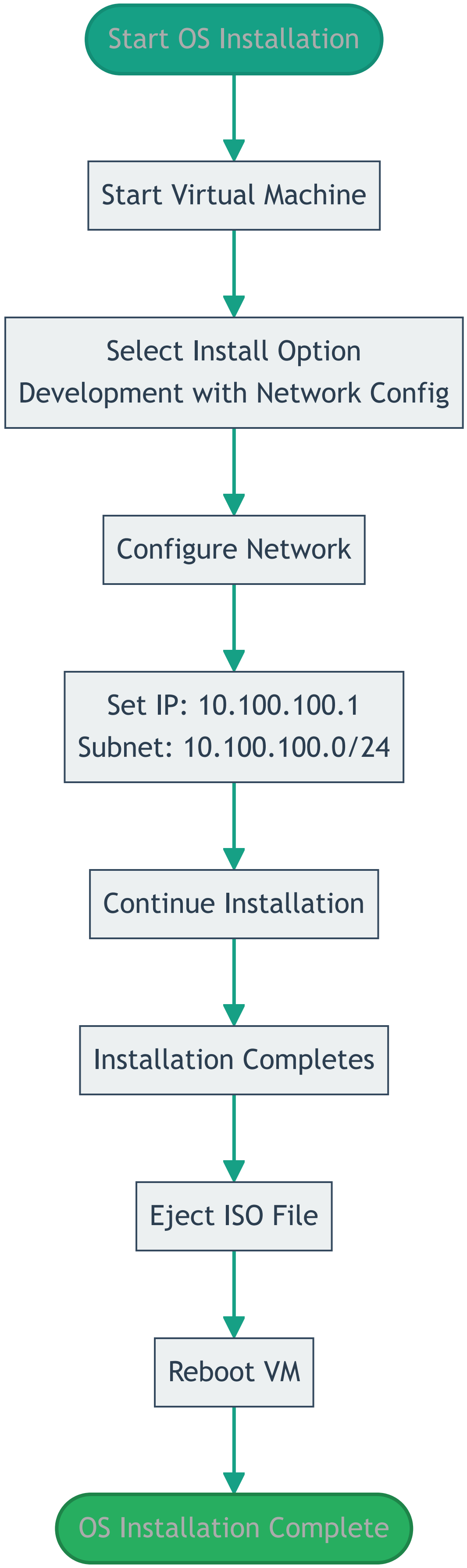
OS Installation Detail

This phase installs and configures the UltraEdge Linux operating system.


Start OS Installation
Start Virtual Machine
Select Install Option
Development with Network Config
Configure Network
Set IP: 10.100.100.1
Subnet: 10.100.100.0/24
Continue Installation
Installation Completes
Eject ISO File
Reboot VM
OS Installation Complete

Key Steps:

- Start the virtual machine for the first time
- Select "Install UltraEdge Instrument: Development with Network Config"
- Configure network interface (eth0) with manual IPv4 settings
- Set IP address to 10.100.100.1 with subnet 10.100.100.0/24
- Complete the installation process
- Eject the ISO file before reboot
- Reboot into the installed system

↑ Back to overview

Next Steps

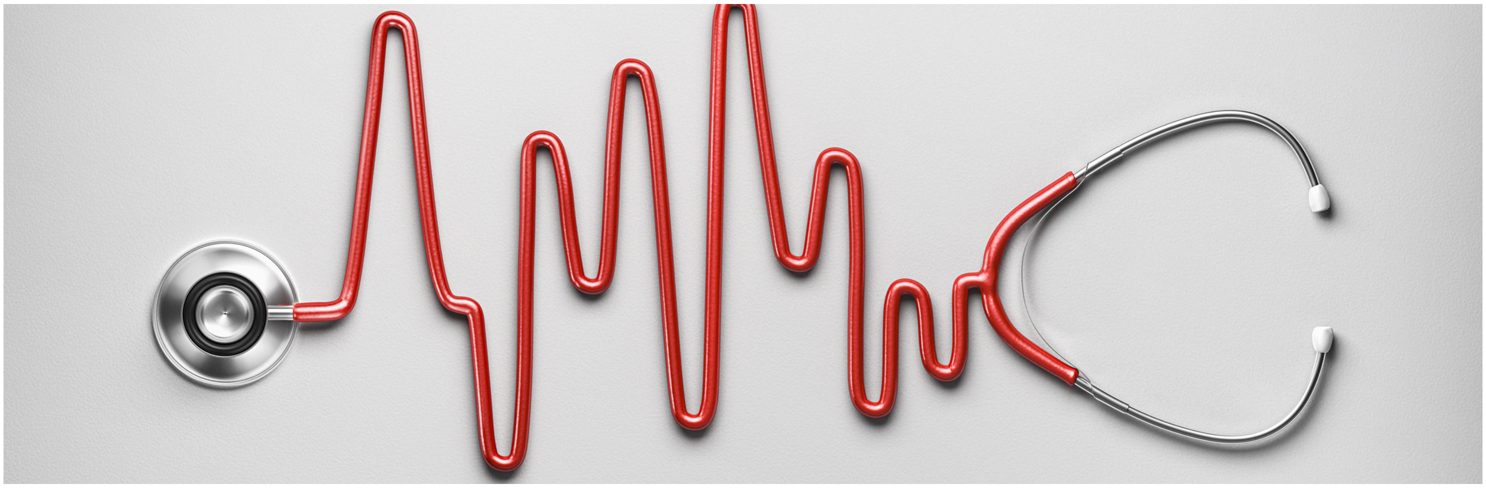
Ready to begin? Follow the detailed step-by-step guides:

- Overview - Introduction to Hyper-V for UltraEdge
- Hyper-V Activation - Enable Hyper-V feature on Windows
- Nested VMs - Enable virtualization for VMs inside VMs
- Installation Guide - Complete installation instructions

This documentation is part of the Teradyne Archimedes Reference Architecture series.

About Hyper-V...

Hyper-V is a native hypervisor developed by Microsoft, available on Windows Server and Windows 10 Pro and Enterprise editions. It allows you to run multiple virtual machines on a single physical host, making it ideal for large-scale, enterprise-level simulations. To create a virtual machine in Hyper-V, first ensure that virtualization is enabled in your BIOS settings. Then, use the Hyper-V Manager to create a new virtual machine, allocate resources like CPU, memory, and storage, and configure network adapters for the VM. You can access the ISO file directly via the Hyper-V configuration interface.



Nested VMs: enabling VMs inside VMs with Hyper-V

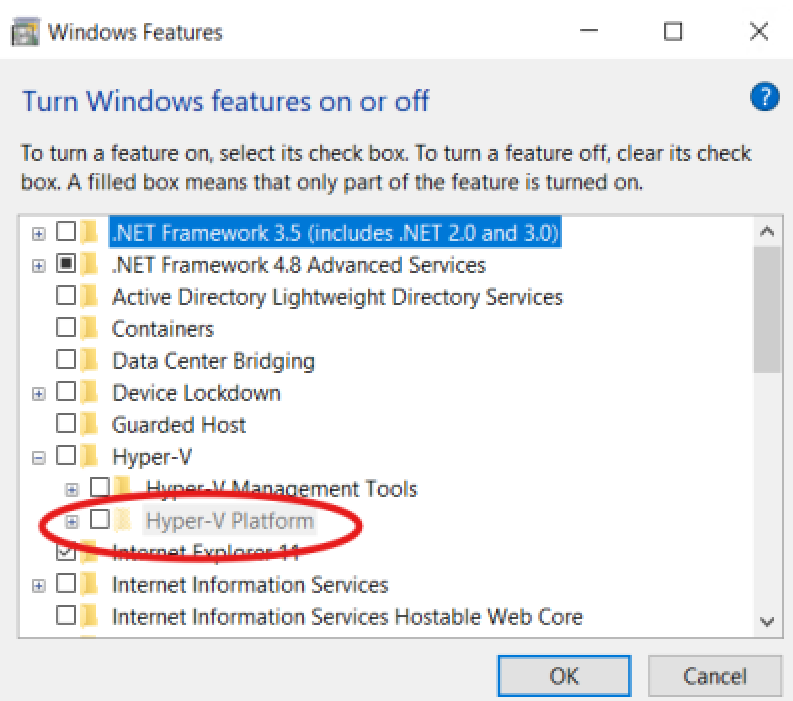
To create virtual machines inside a Hyper-V virtual machine, you need to enable processor virtualization features.

The virtual machine needs to be turned off and the following PowerShell command must be issued as an administrator.

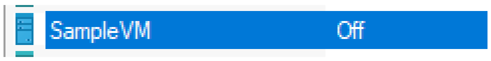
```
Set-VMProcessor -VMName <VMName> -ExposeVirtualizationExtensions $true
```

where is the name (not the path) of your virtual machine as seen in Hyper-V Manager.

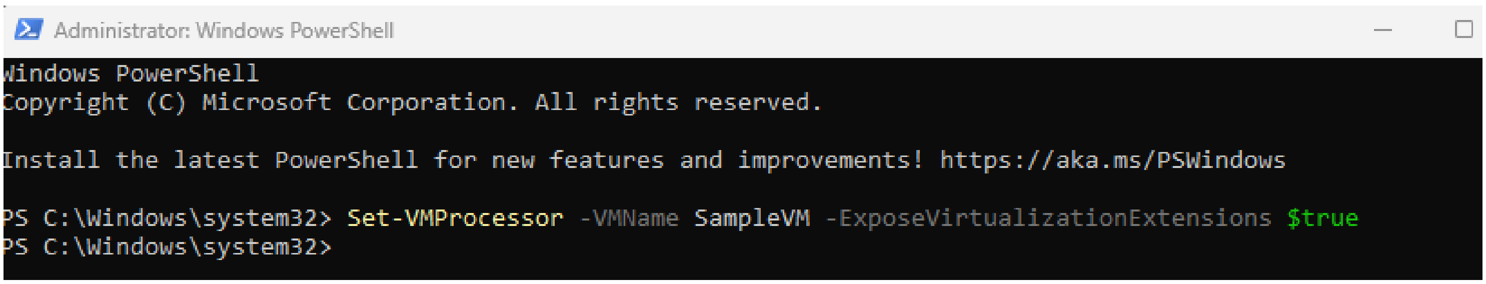
For example, I created a Hyper-V virtual machine called SampleVM on my host PC. Inside this VM, I want to enable Hyper-V to created nested virtual machines. In the “Turn Windows features on or off”, I get a grayed out option which prevents me from using Hyper-V virtual machines inside SampleVM:



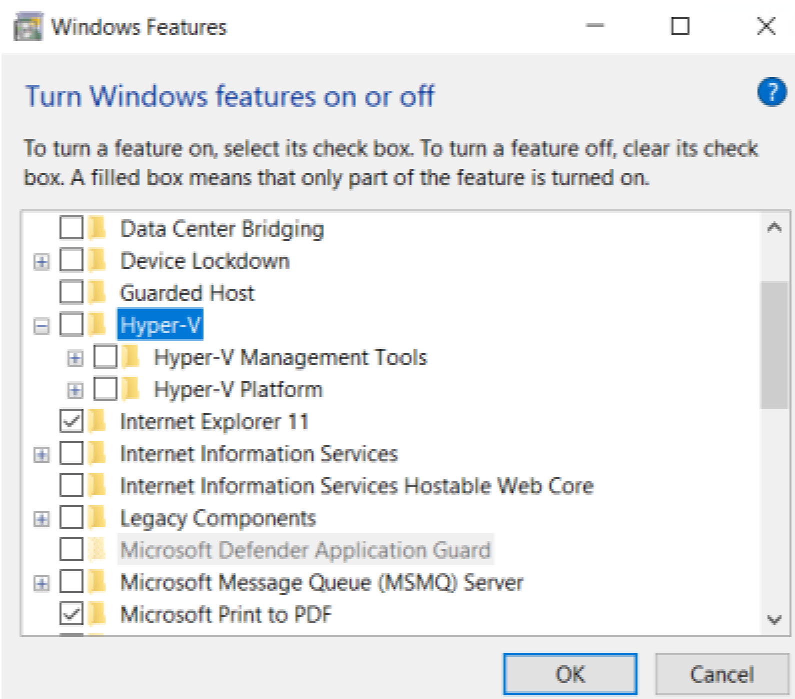
Let's shutdown the virtual machine.



Now, on my host PC, inside a PowerShell console started as an administrator, I issue the following command. As you can see, there is no message whatsoever in case of success:



I can now turn SampleVM back on and I can see that Hyper-V is fully available now:



Hyper-V activation

Hyper-V is a Microsoft Windows feature that needs to be enabled in order to use Hyper-V virtual machines.

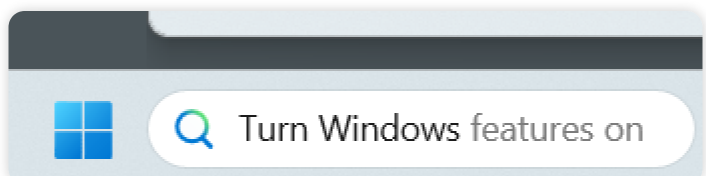
The following steps describe how to enable Hyper-V on a Windows host.

Note that the scripts take care of that activation if it was not done previously.

If you need to enabled Hyper-V within a Hyper-V virtual machine (nested VMs), the host VM needs to have a virtual processor with virtualization features enabled. Go to this page for information: [Nested VMs]

Step 1

In the Windows task bar, search for “Turn Windows features on or off”

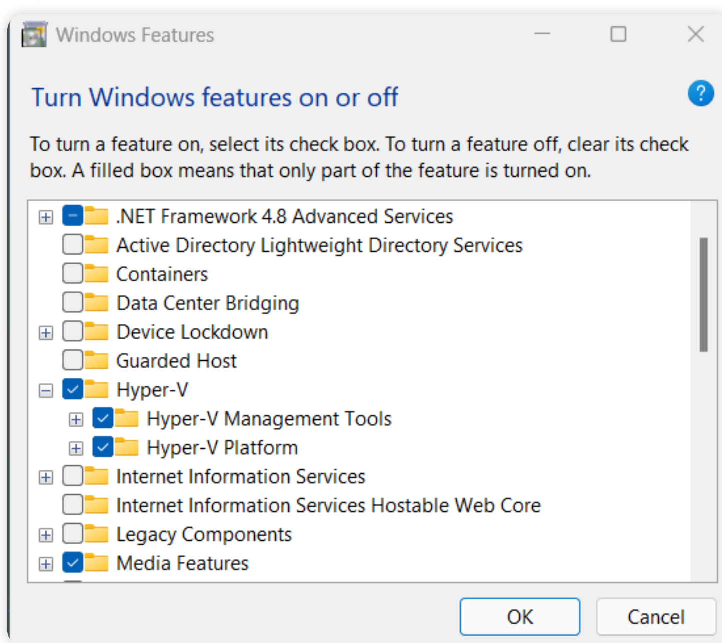


Step 2

In the window that appears, scroll down to Hyper-V and make sur to check all the Hyper-V features. Note that these might have already been checked. In this case, you are all set.

It is recommended to reboot the machine after activating this feature.

Note: if you are also using VMWare or VirtualBox, specific steps might be required to enable both environments to cohabitate. Please refer to the VMWare or VirtualBox documentation.



Installation

The following sections describe how to install a Hyper-V UltraEdge simulator.

All steps are listed from the creation of a new virtual machine until the boot to the login prompt.

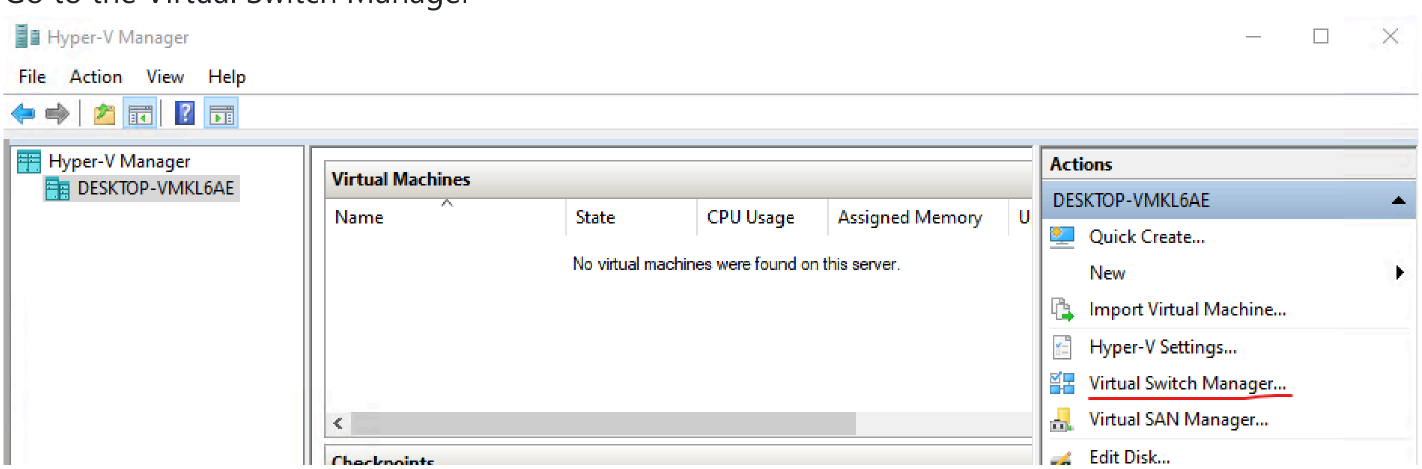
Prerequisite: the UltraEdge Emulator ISO file must be downloaded from eKnowledge.

On eKnowledge, search for `UltraEdge ISO` to get to the download page. The image usually comes as a compressed file (7z and zip), please uncompress it to obtain the ISO file. The uncompressed ISO file should be around 2GB.

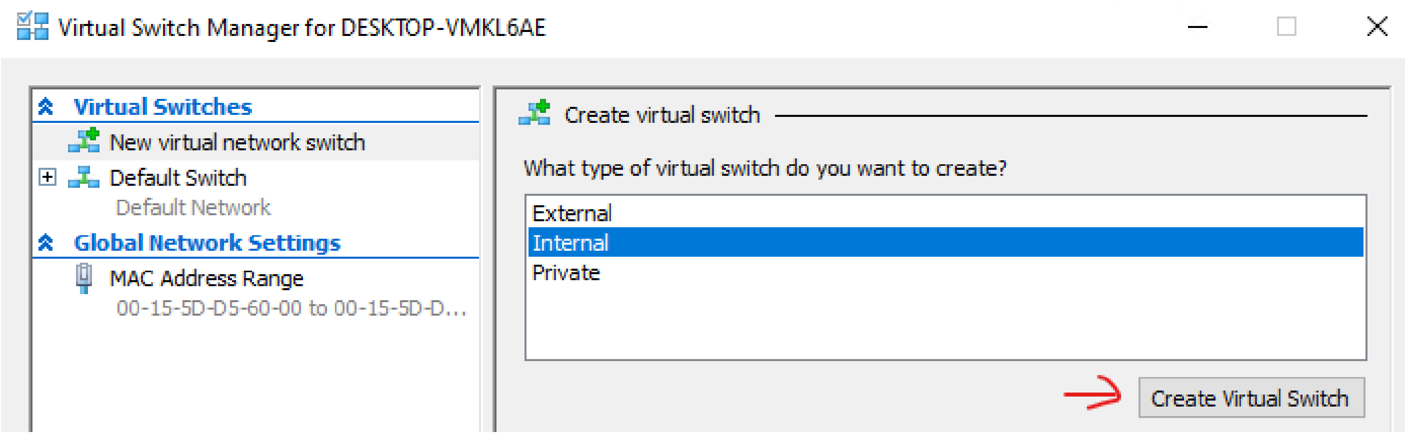
Network switch setup

A network switch makes the network setup simpler. Create a new Virtual Switch for the UltraEdge VM.

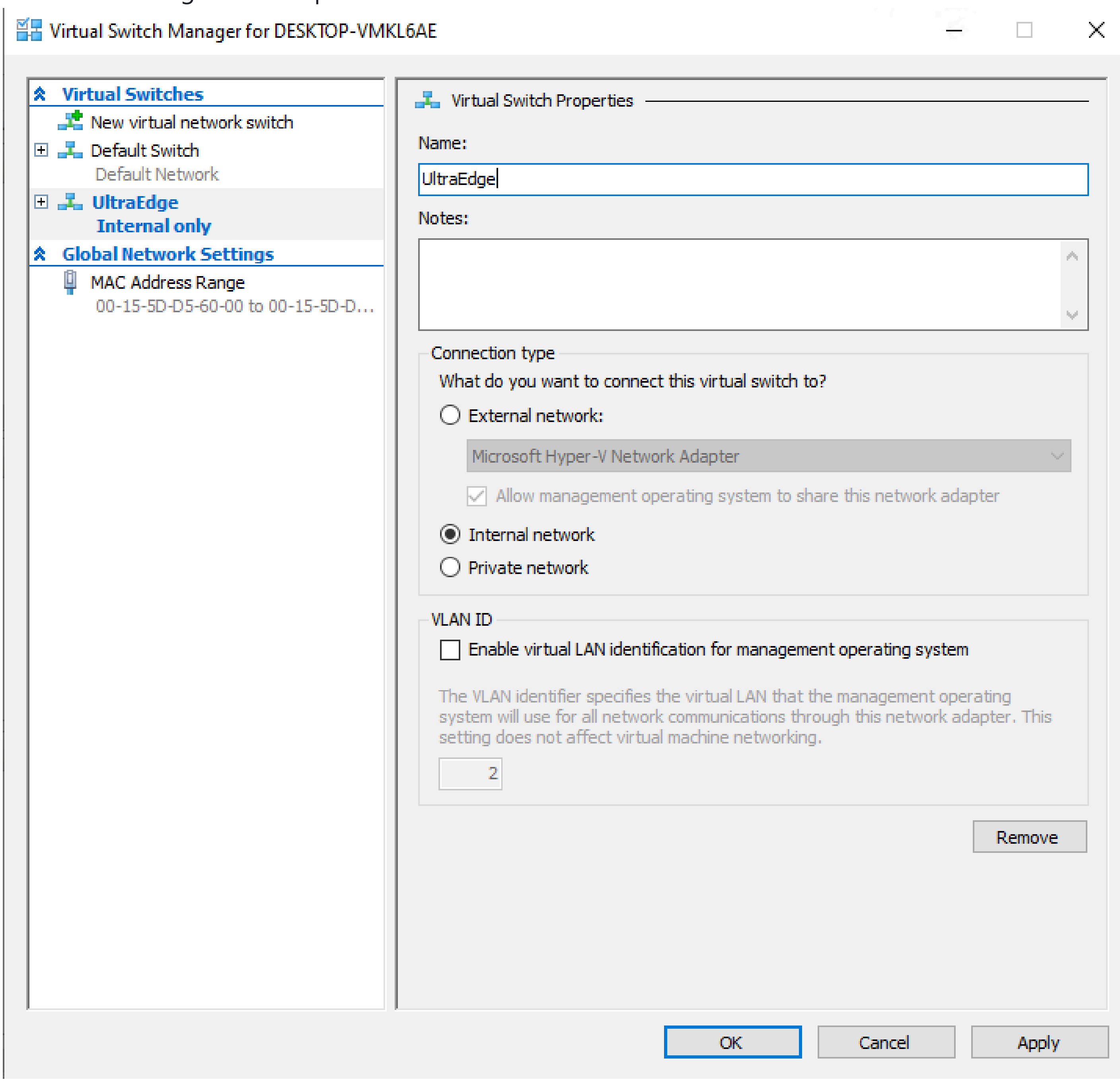
- Go to the Virtual Switch Manager



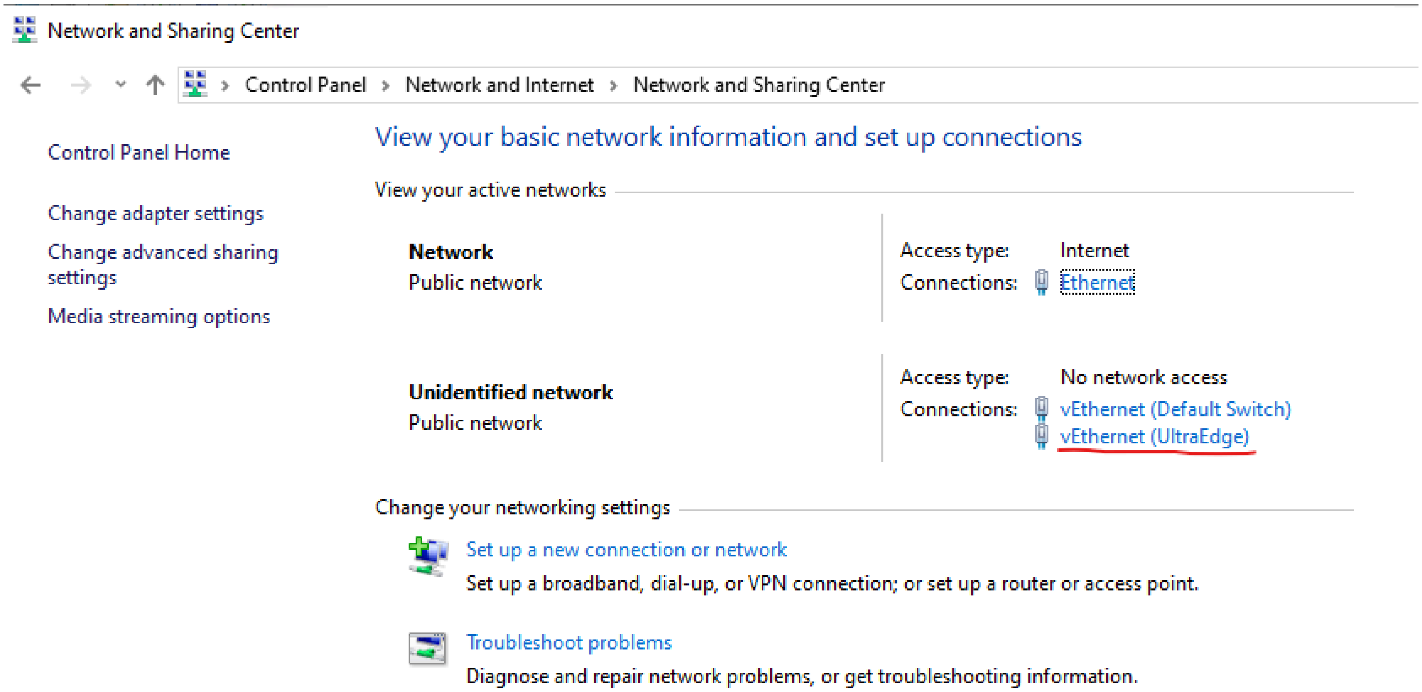
- Create a new Internal Virtual Switch



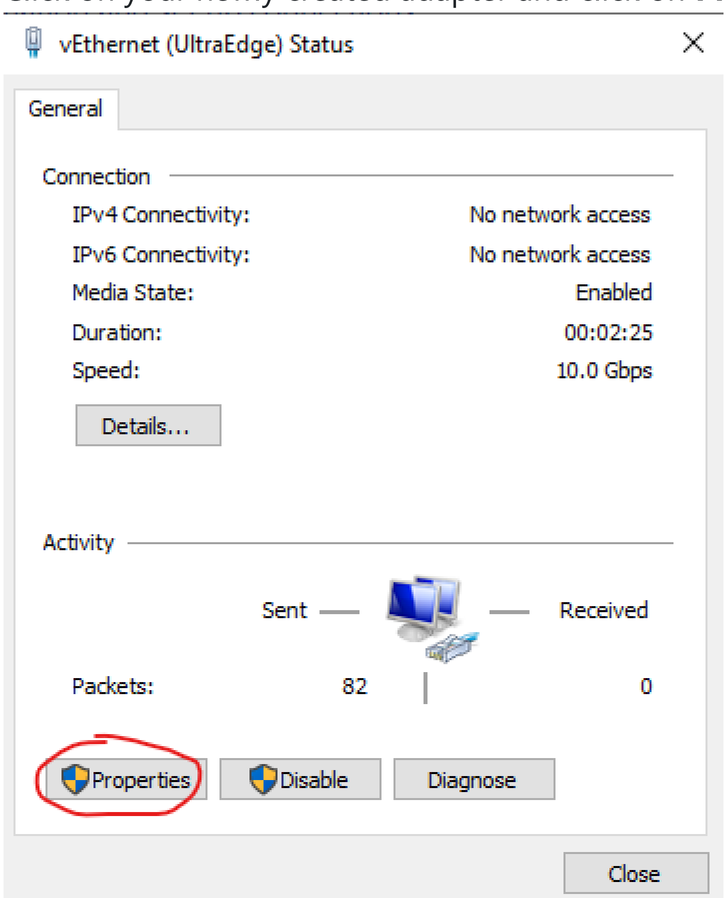
- Name it UltraEdge for example



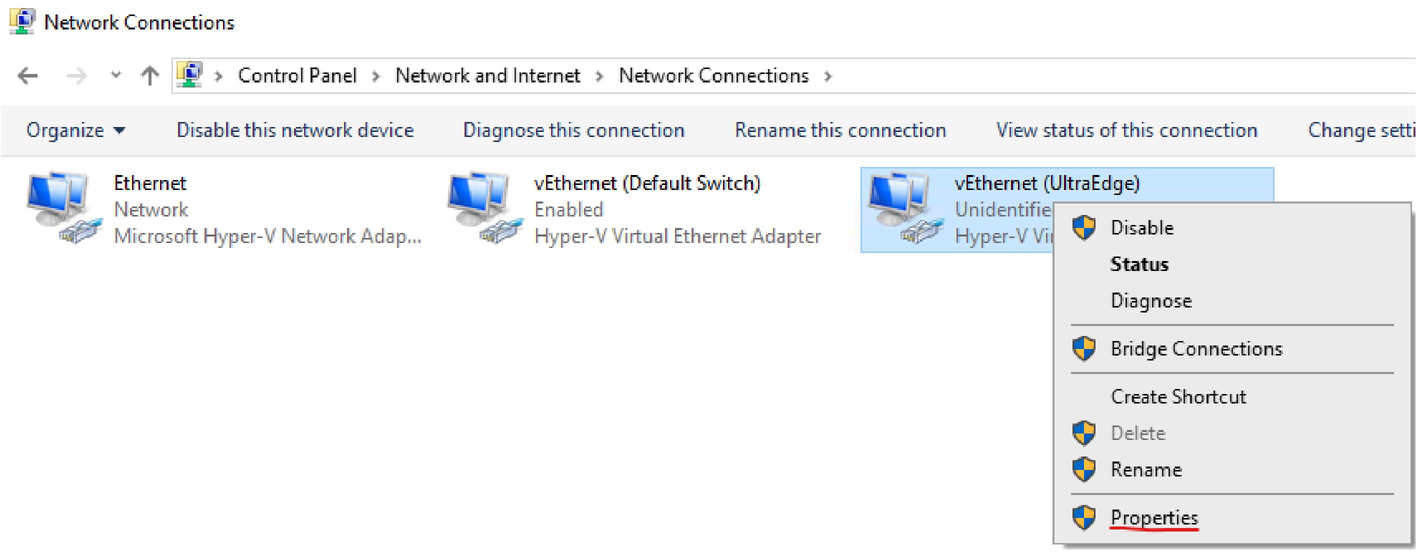
- Go to the Windows **Network and Sharing Center**



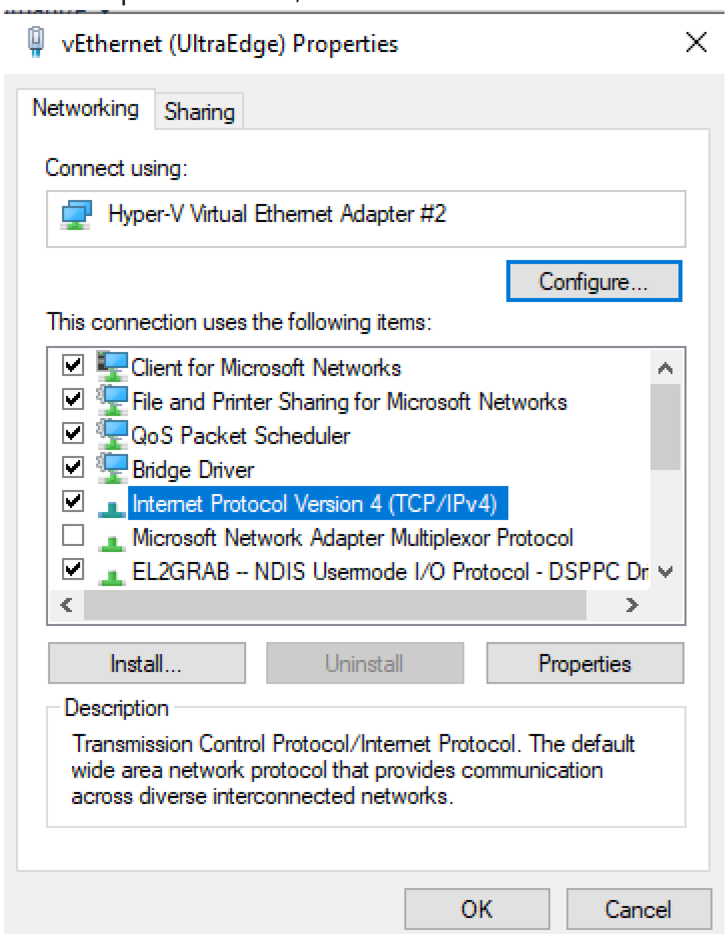
- Click on your newly created adapter and click on **Properties**



- Alternatively, go to **Change adapter settings** on the left side, right-click on your newly created adapter and select Properties

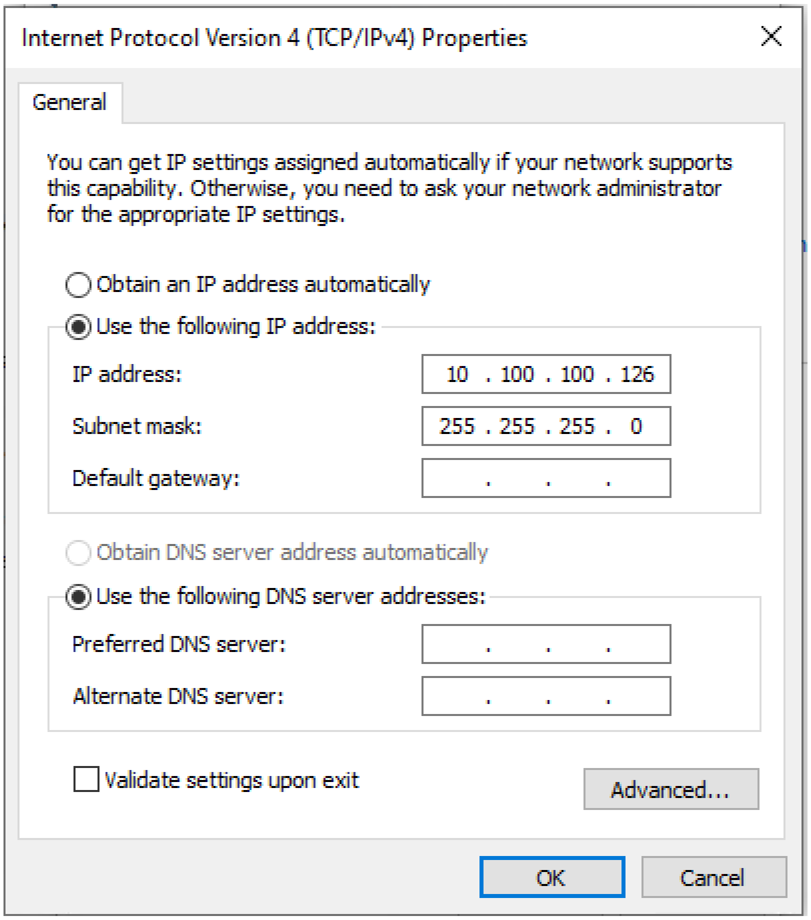


- On the network adapter window, select **Internet Protocol Version 4 (TCP/IPv4)** and click on **Properties**



- Fill the **IP address** and **Subnet mask** as shown here:
  - IP address = 10.100.100.126

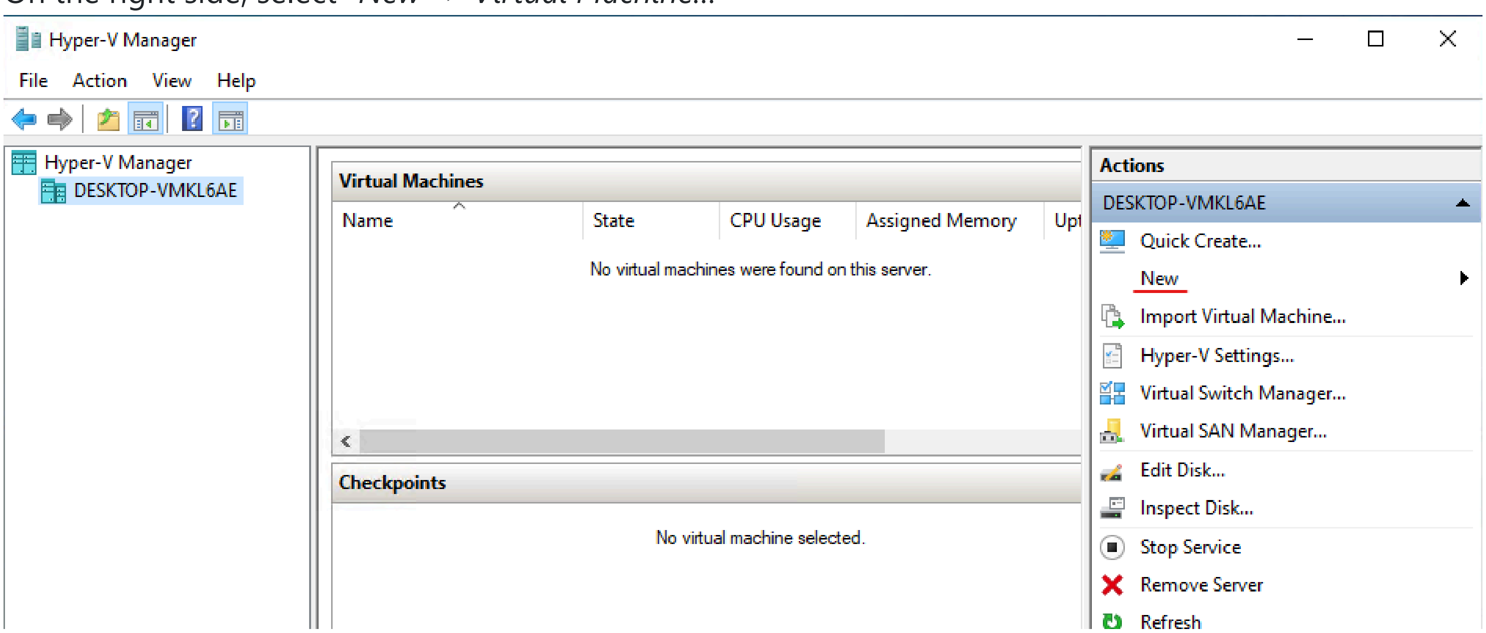
- Subnet mask = 255.255.255.0

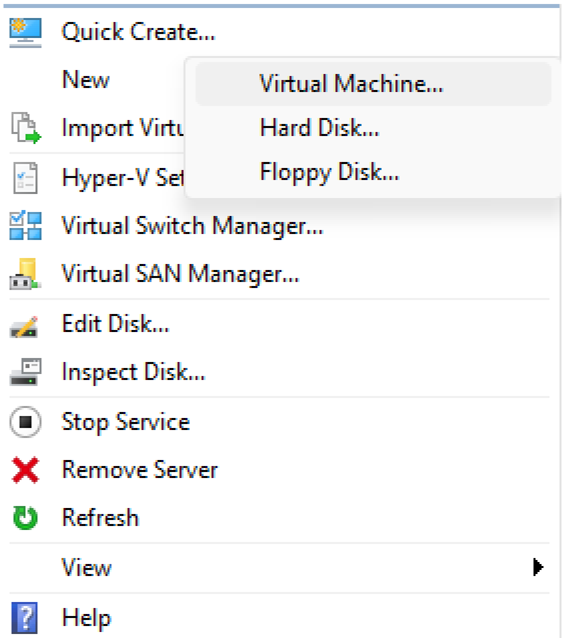


Hyper-V Virtual Machine creation and setup

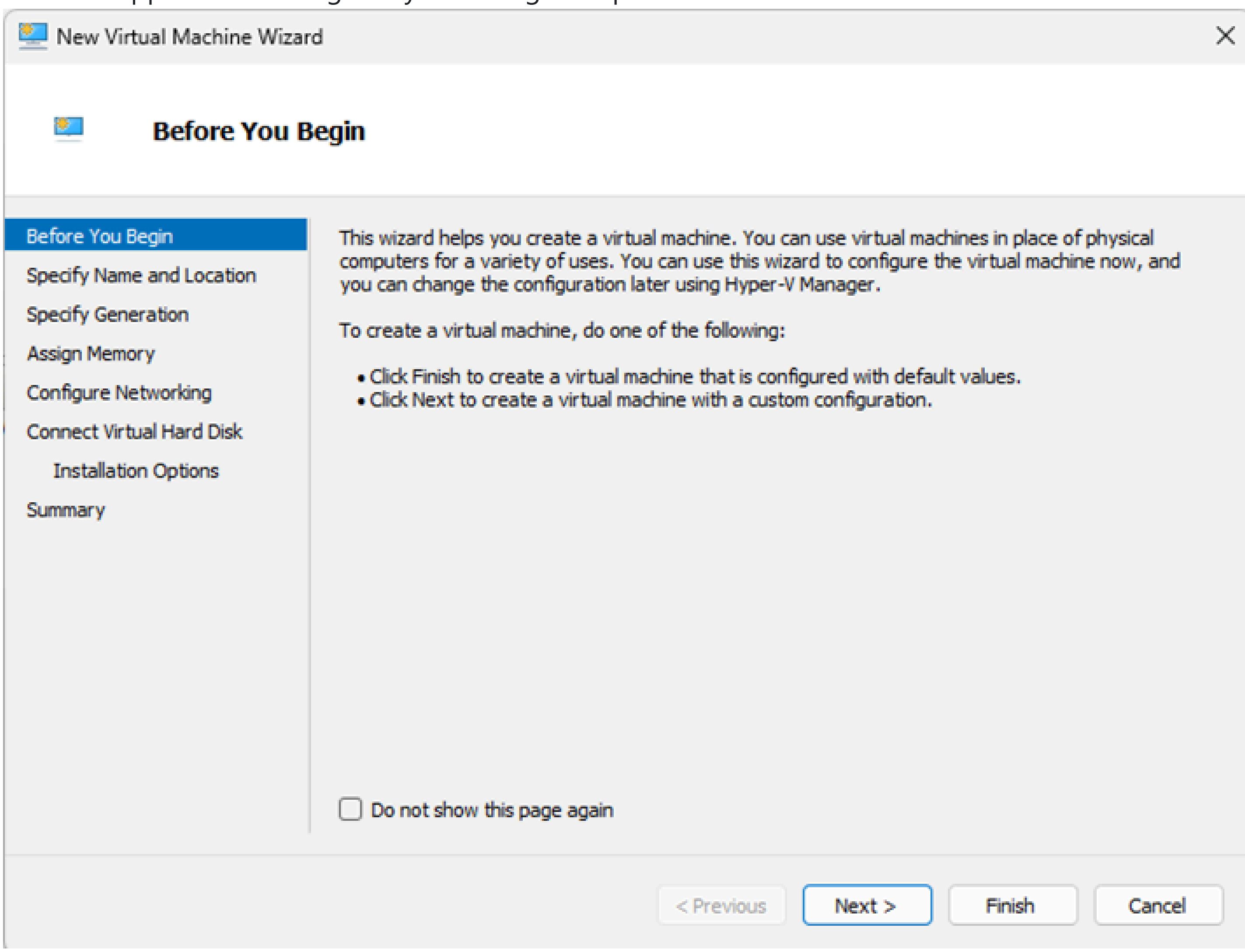
Open the Hyper-V manager.

On the right side, select *"New -> Virtual Machine..."*

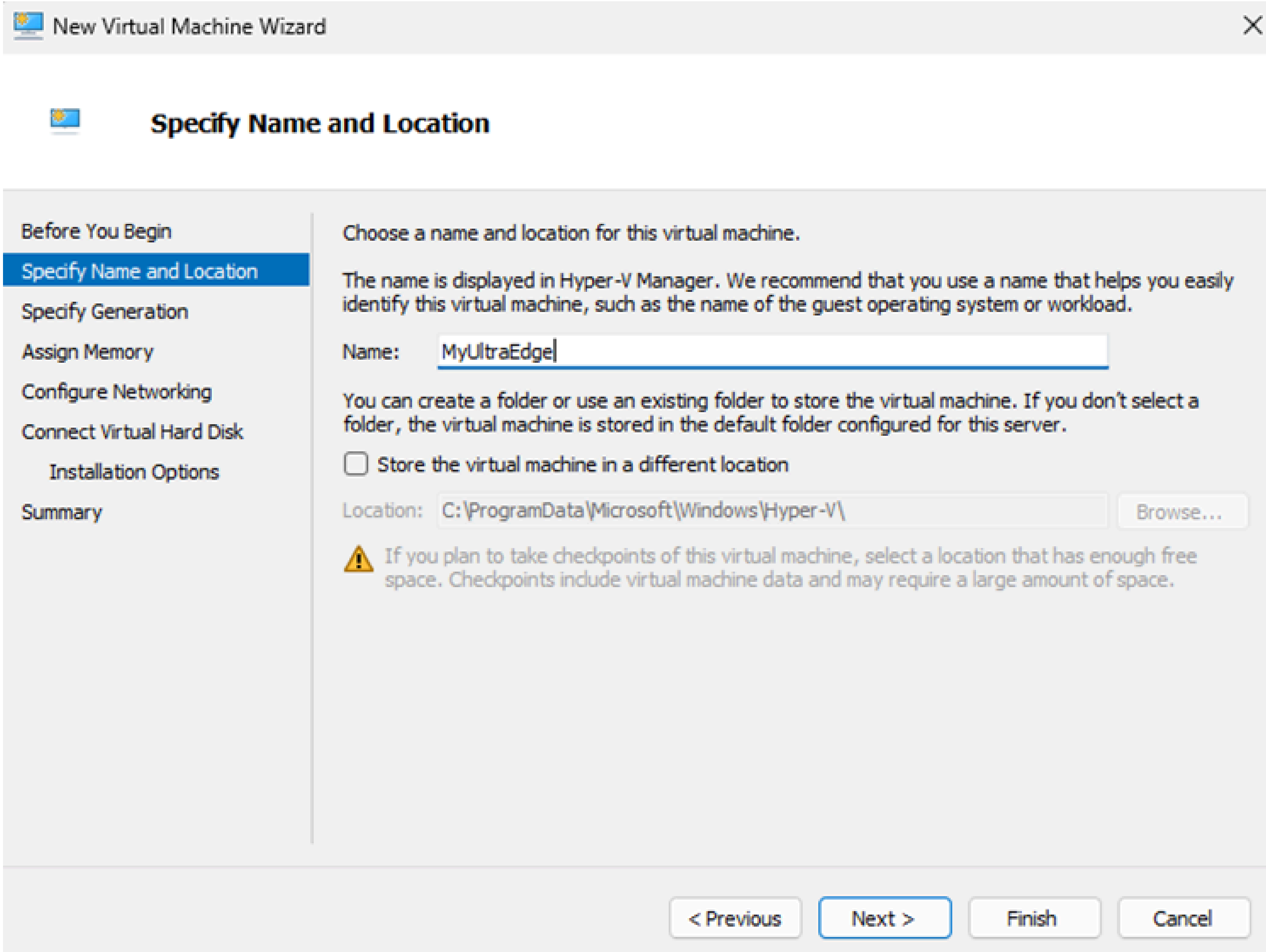




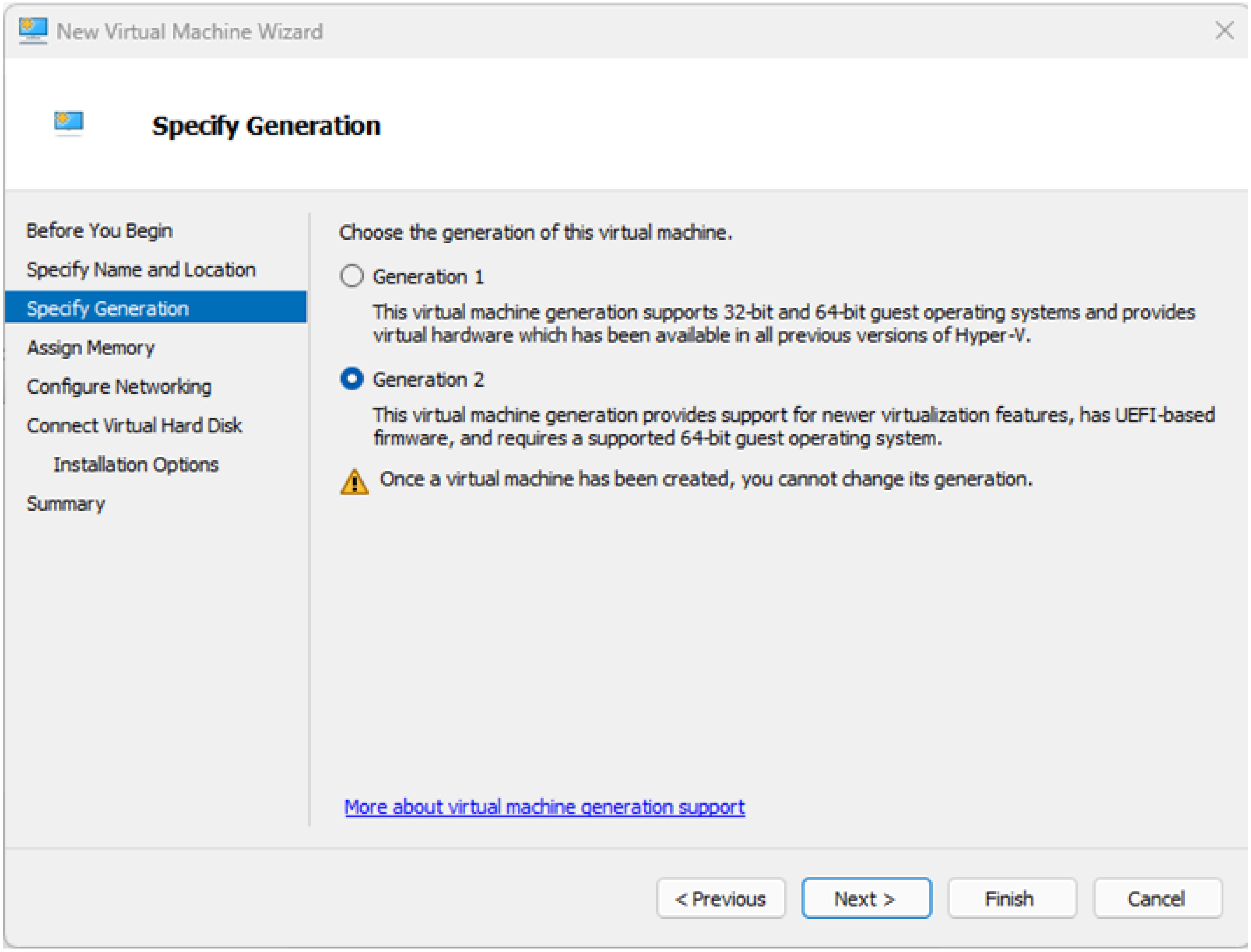
A wizard appears that will guide you through the process. Click **Next>**



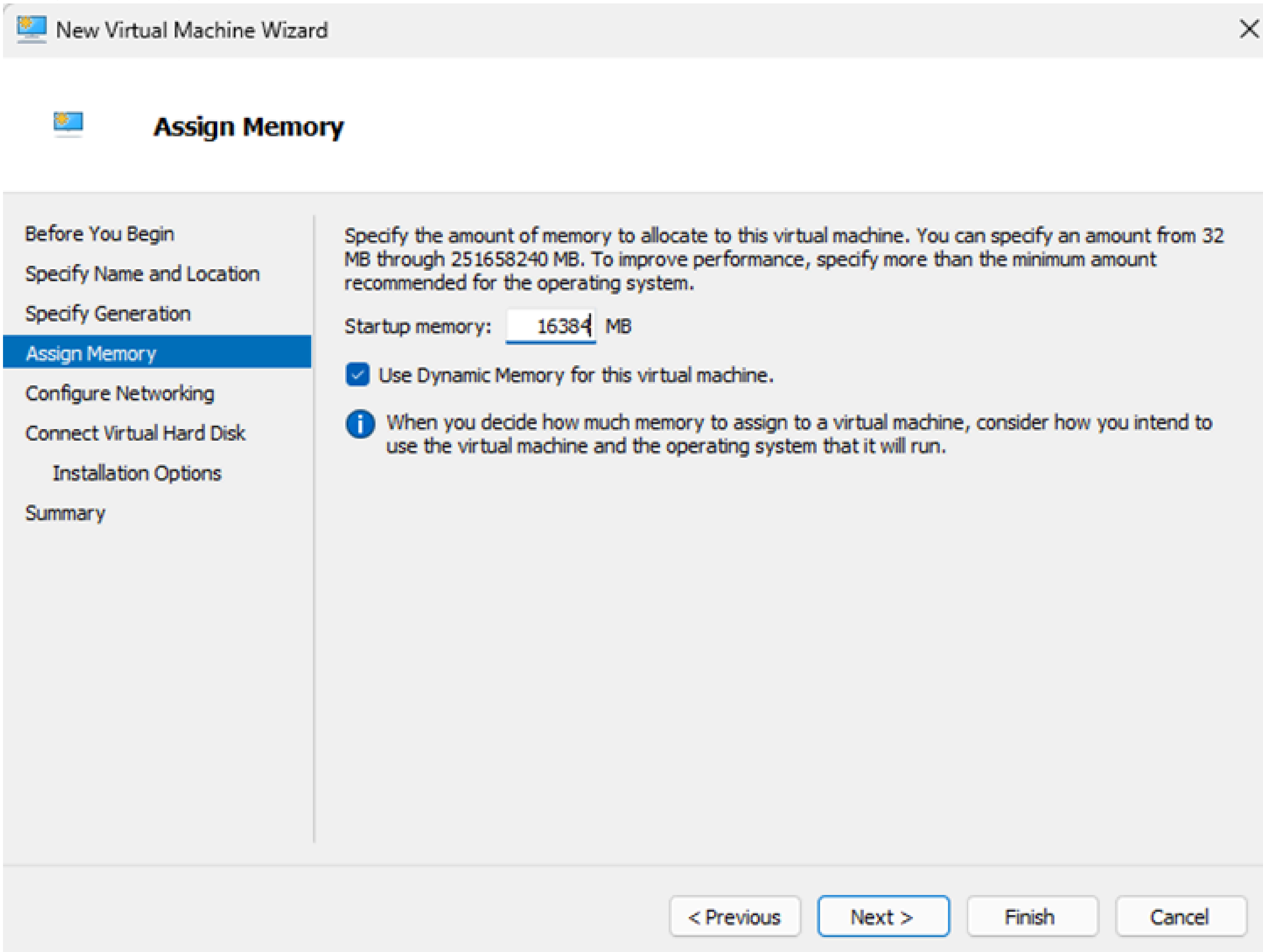
Specify the name of the virtual machine (here MyUltraEdge). **Click Next>**



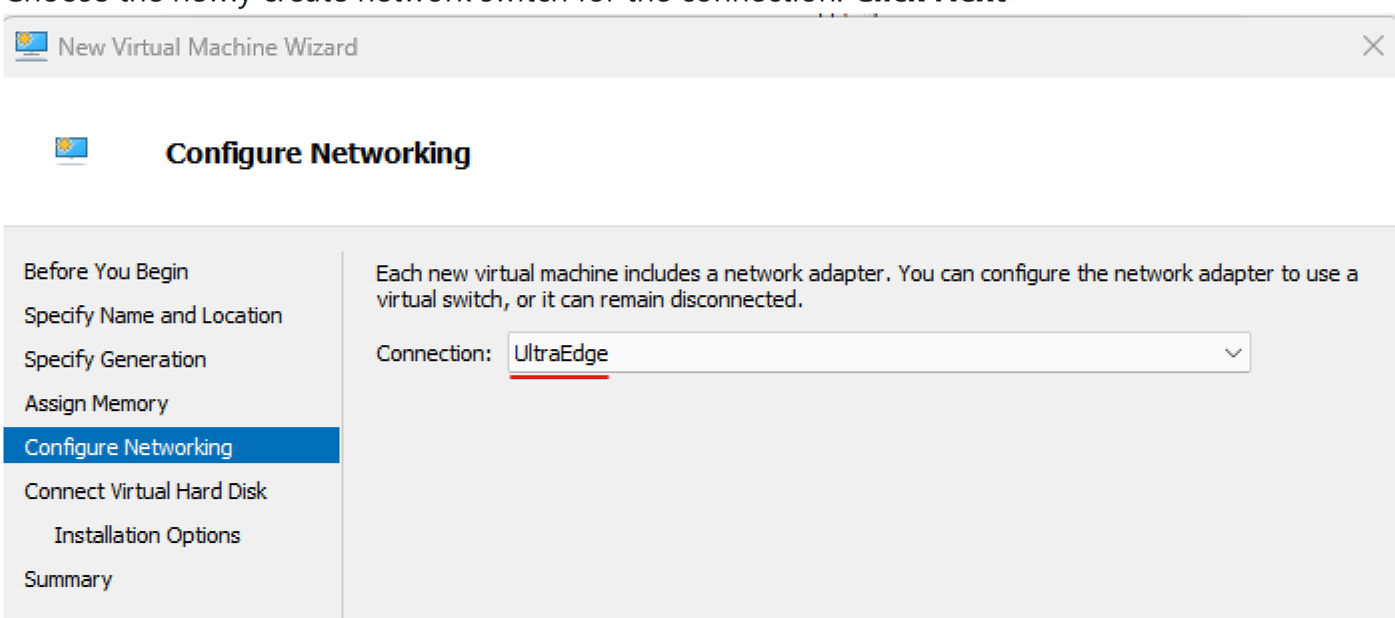
The UltraEdge Linux distribution requires UEFI boot. Select **Generation 2 Click Next>**



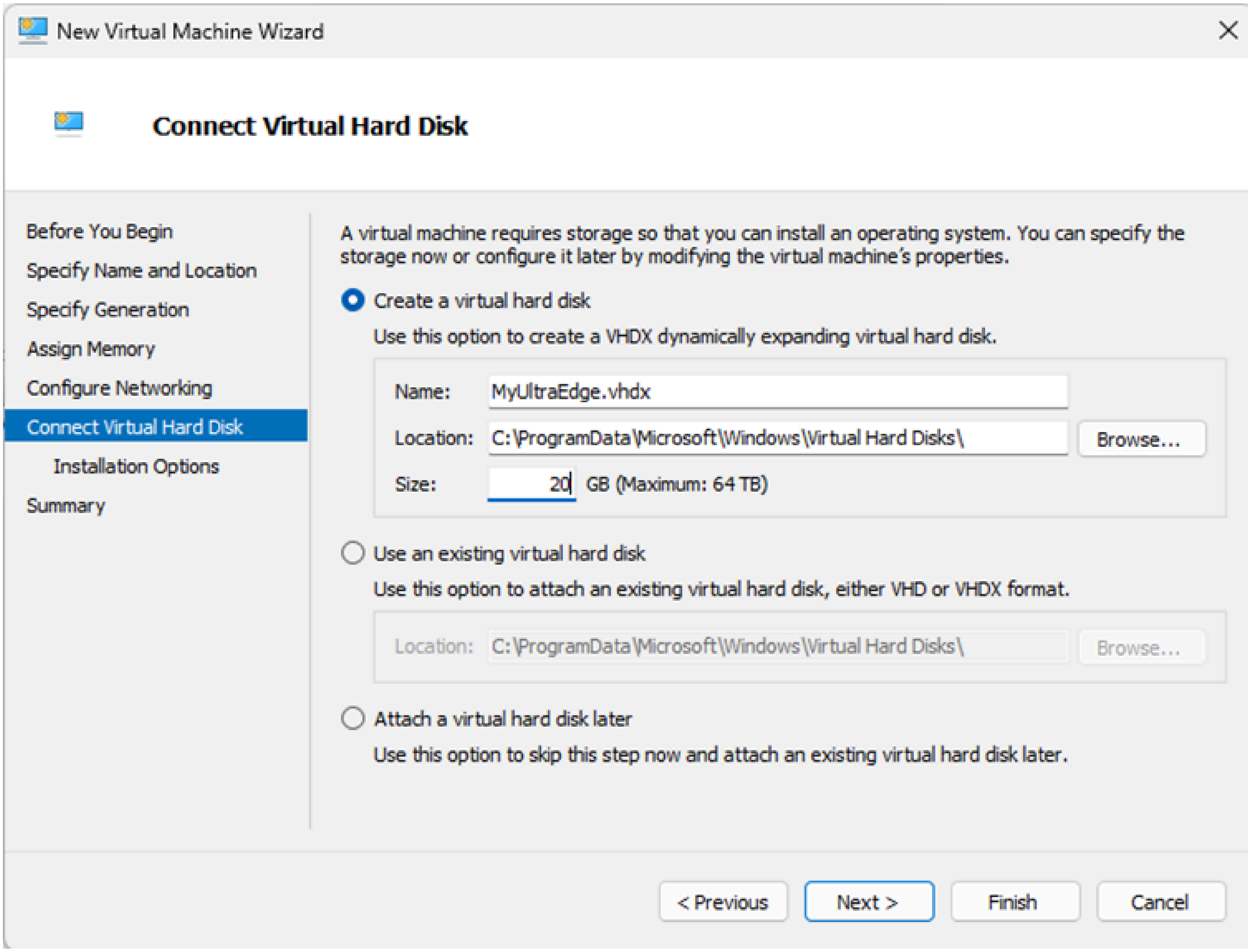
Assign memory with a minimum of **4096MB**. **Click Next>**



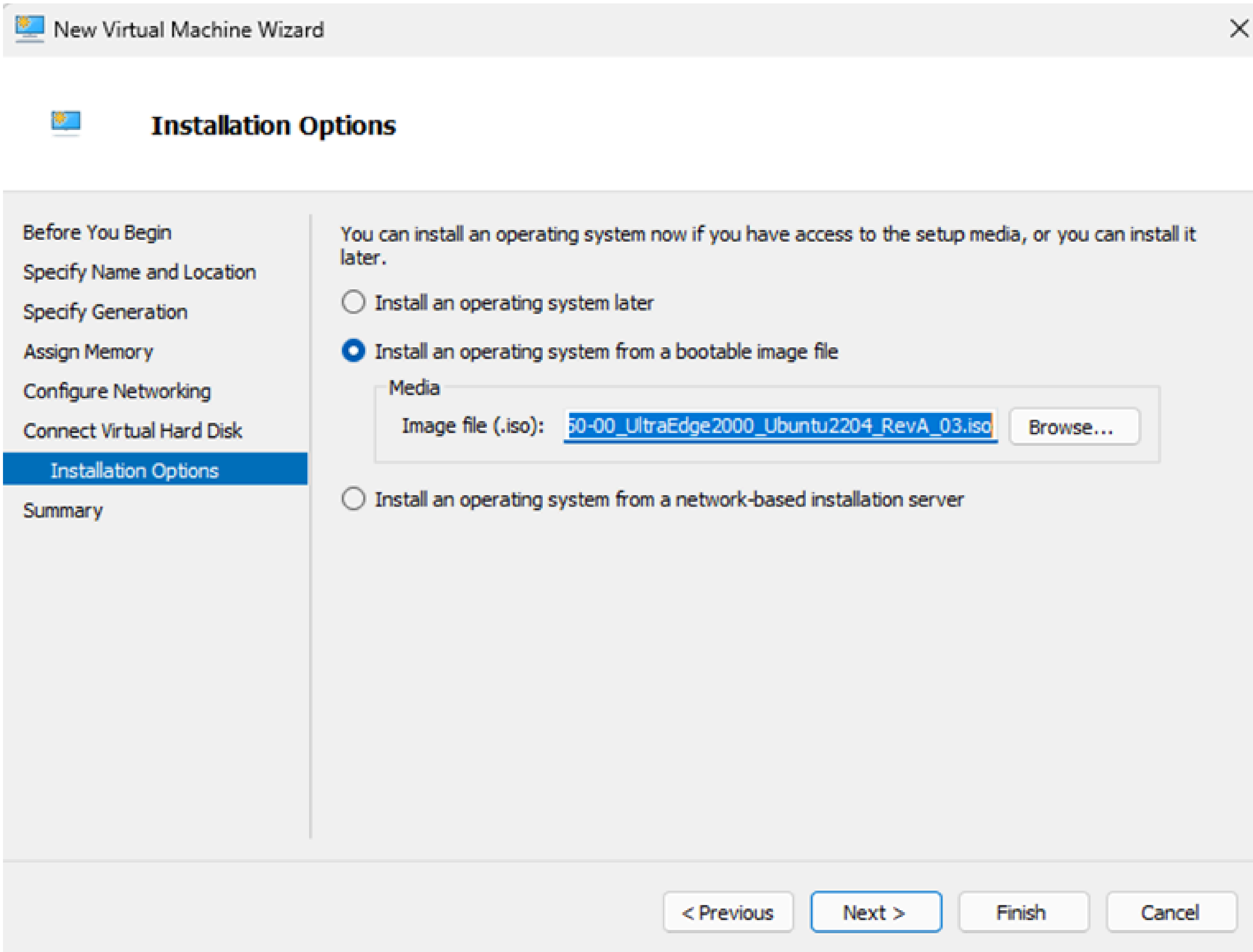
Choose the newly create network switch for the connection. **Click Next>**



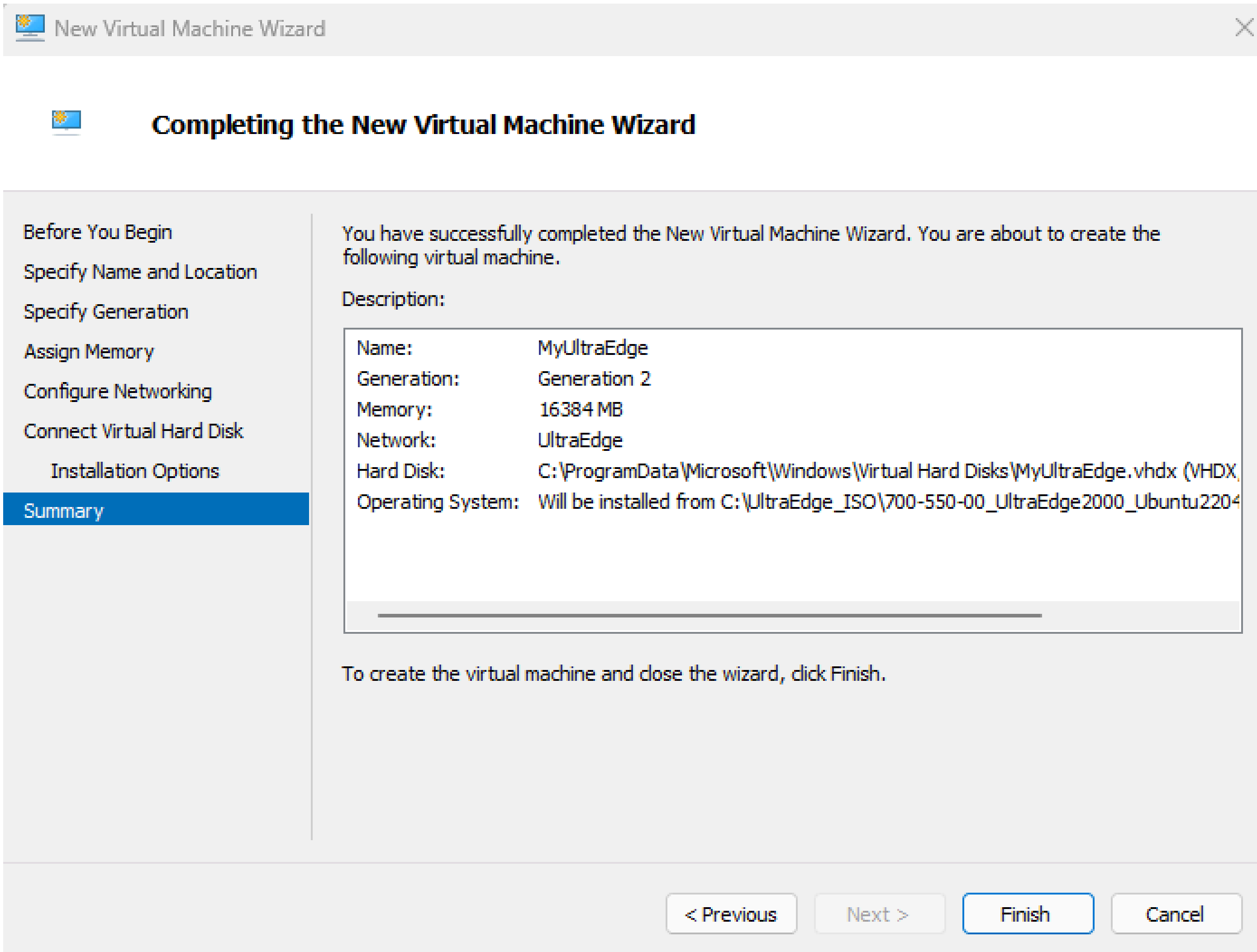
Create a virtual hard disk of size 20GB. **Click Next>**



Select `Install an operating system from a bootable image file` and browse to select the ISO file you downloaded. **Click Next>**



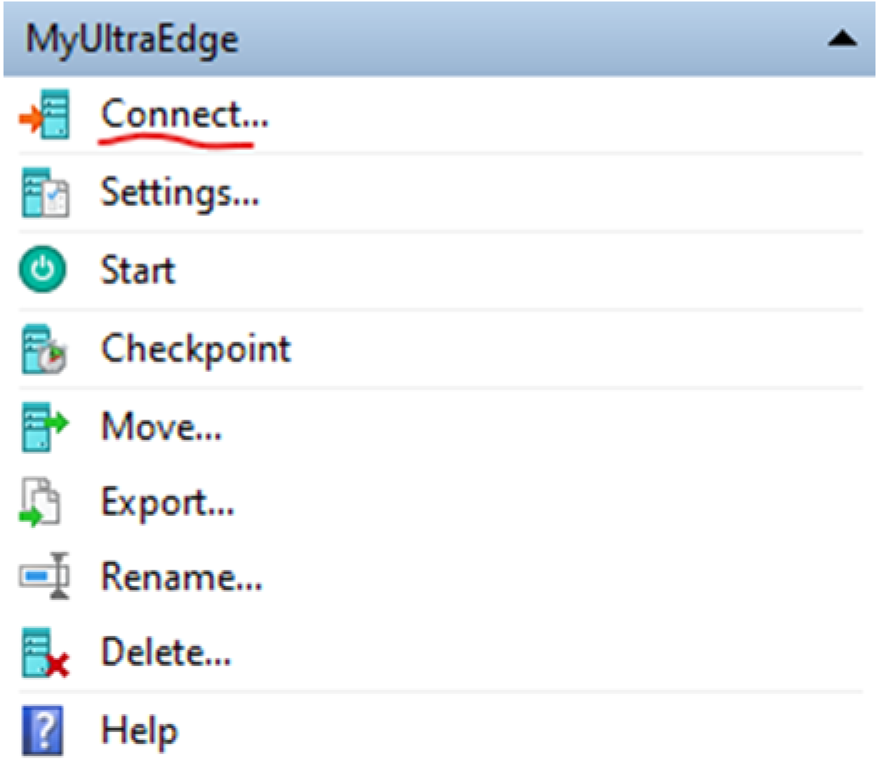
Review your selections and Finish Alternatively, use **<Previous** to go back in the wizard and modify a selection.

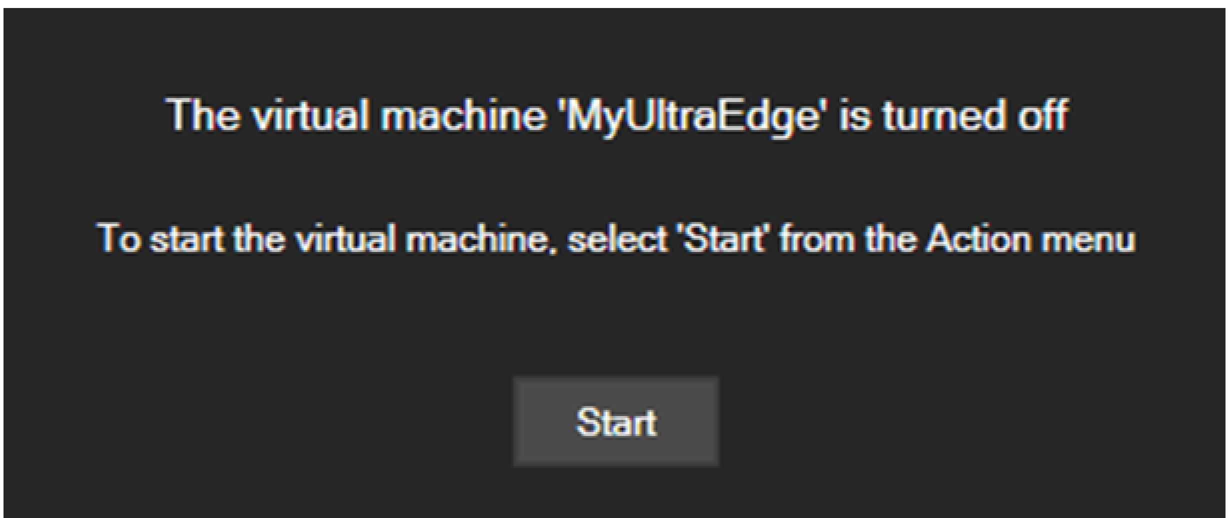


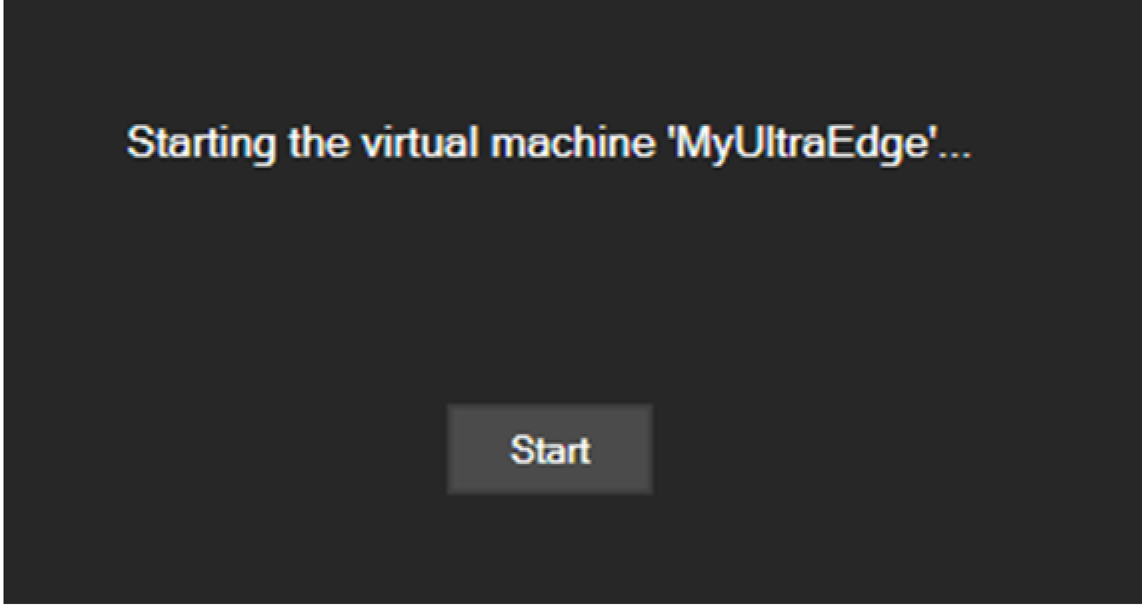
First Start

Once the wizard is finished, you can connect to your new UltraEdge virtual machine. The system has not been installed yet. This is what needs to be done now. By default, the machine is turned off. Click on

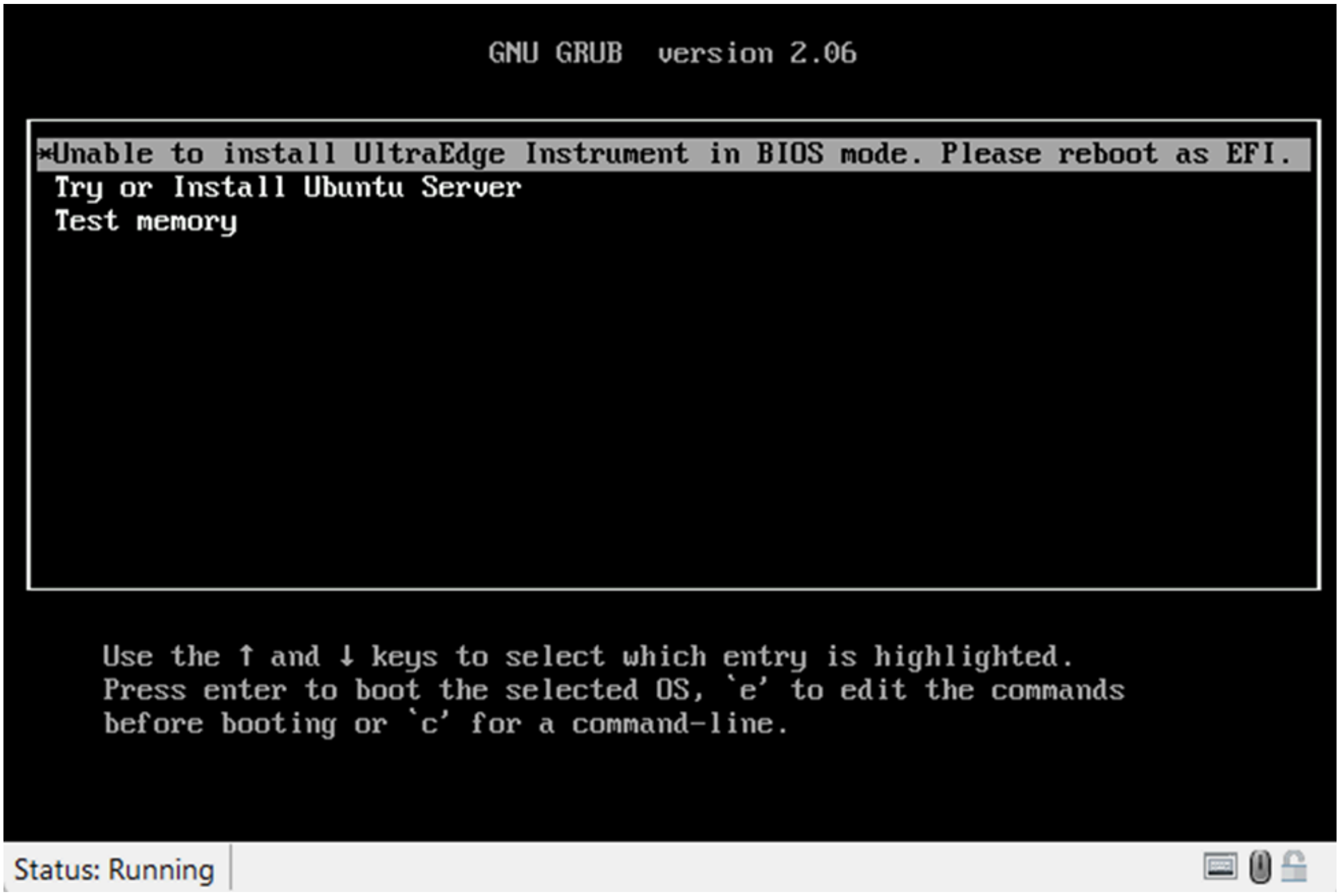
Start.


MyUltraEdge
Connect...
Settings...
Start
Checkpoint
Move...
Export...
Rename...
Delete...
Help

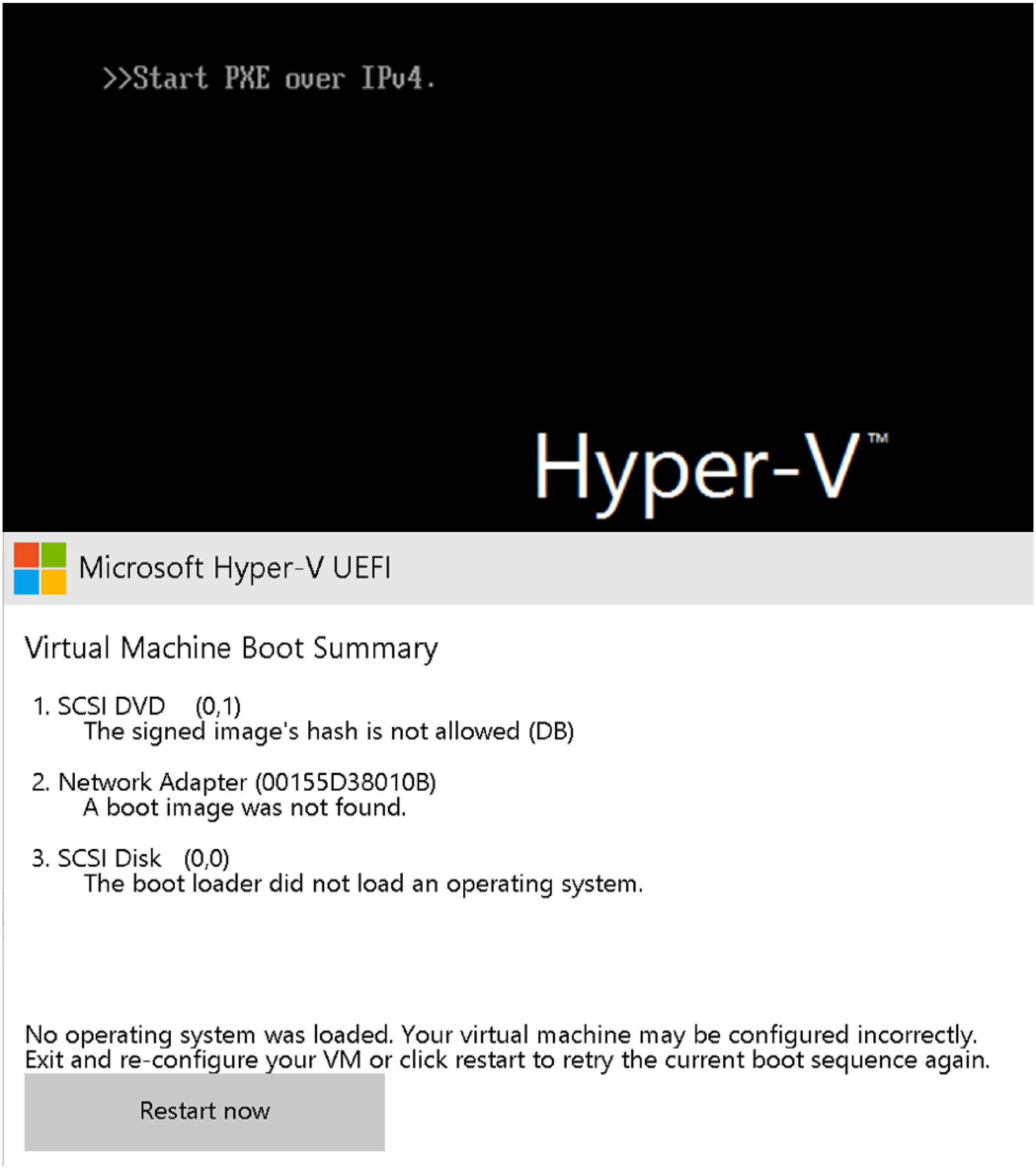

The virtual machine 'MyUltraEdge' is turned off
To start the virtual machine, select 'Start' from the Action menu
Start


Starting the virtual machine 'MyUltraEdge'...
Start

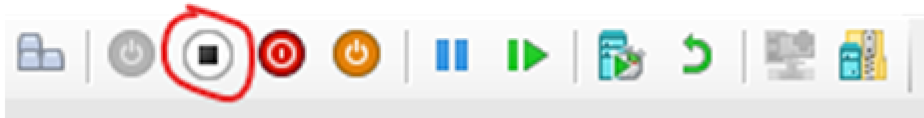
- **Note**: if the Virtual Machine was configured with Generation 1, the following screen will appear.
    - In this case, the easiest course of action is to recreate the virtual machine from scratch making sure that **Generation 2** is selected.

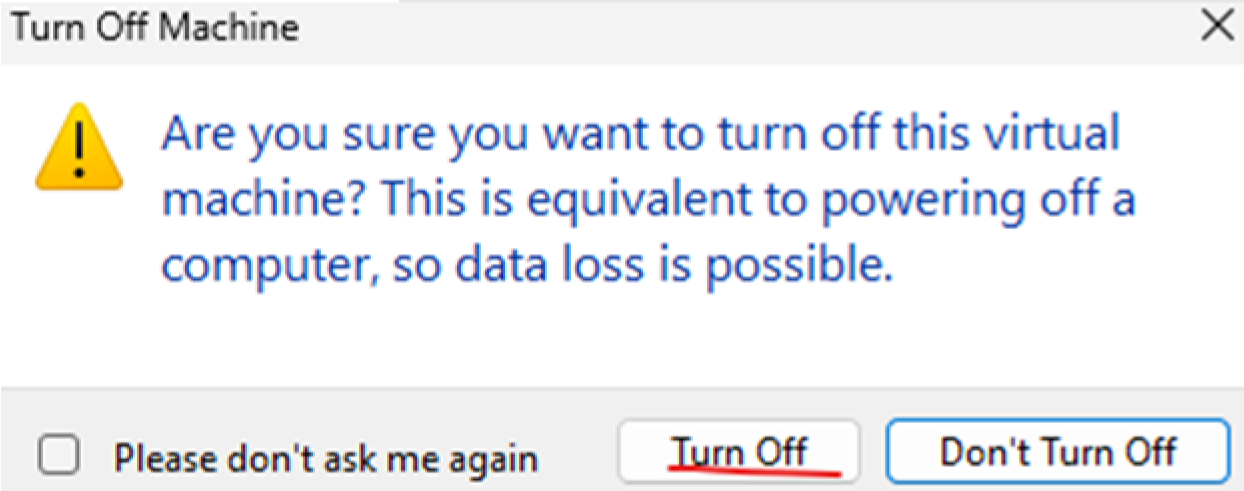


- At first boot, the following screens may appear. The system is not able to start. This is expected, an additional setting needs to be set.


>>Start PXE over IPv4.
Hyper-V™
Microsoft Hyper-V UEFI
Virtual Machine Boot Summary
1. SCSI DVD (0,1)
The signed image's hash is not allowed (DB)
2. Network Adapter (00155D38010B)
A boot image was not found.
3. SCSI Disk (0,0)
The boot loader did not load an operating system.
No operating system was loaded. Your virtual machine may be configured incorrectly.
Exit and re-configure your VM or click restart to retry the current boot sequence again.
Restart now

- Turn off the virtual machine





Security

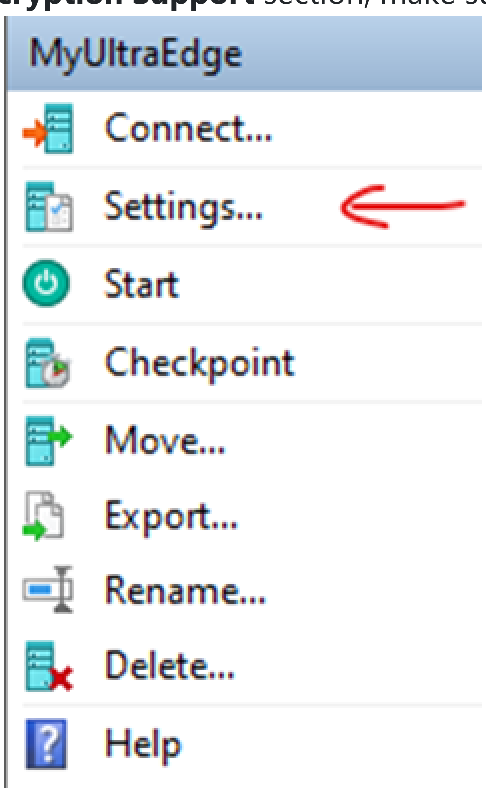
Go to the **Settings** for this virtual machine. Select the **Security** subsection in the **Hardware** section. Make sure the **Enable Secure Boot** is checked (it should be checked by default).

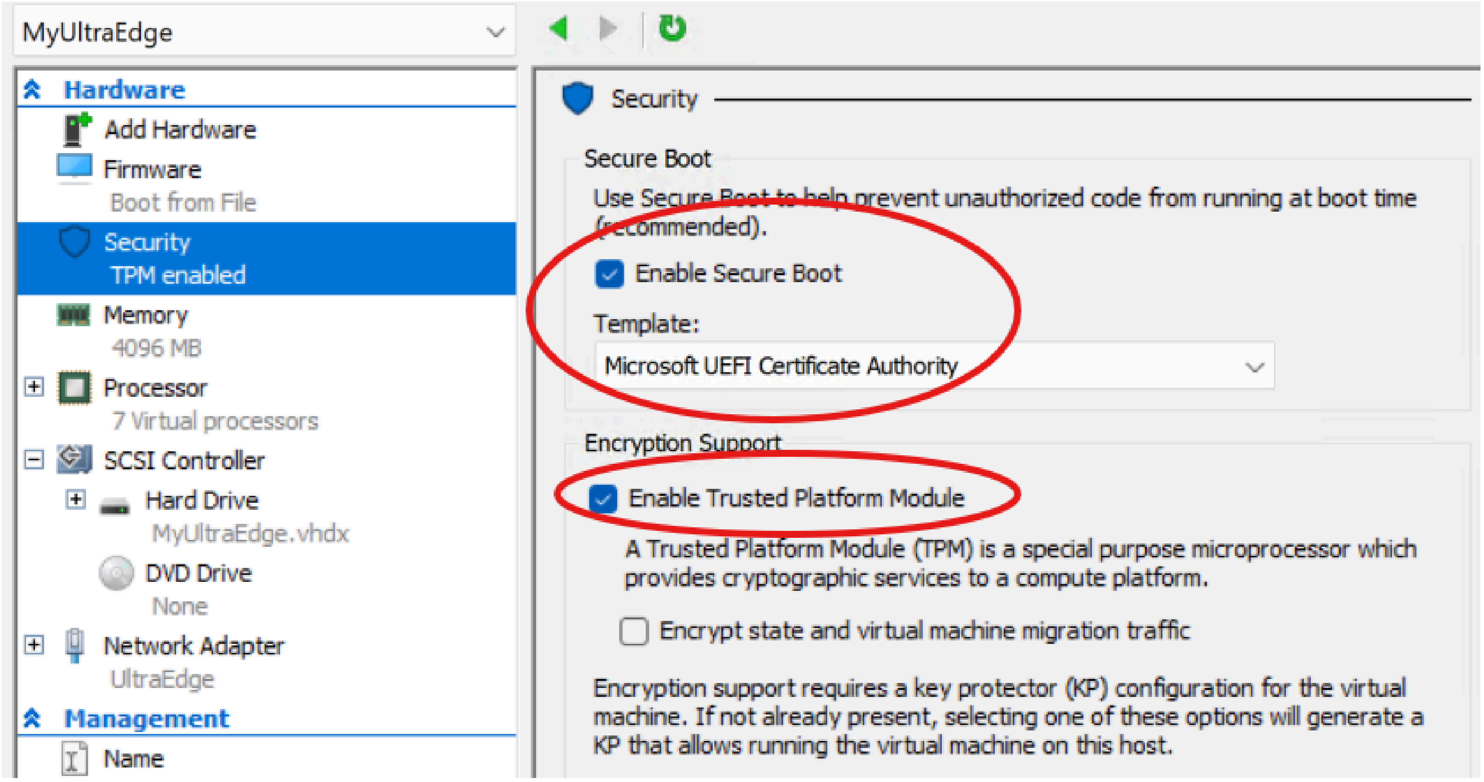
Make sure the selected Template is **Microsoft UEFI Certificate Authority.** This is for Linux guest systems.

The default of this option is *Microsoft Windows*, which does not work with the UltraEdge Linux guest.

It is also the right time to enable the **Trusted Platform Module** that allows for secure connection.

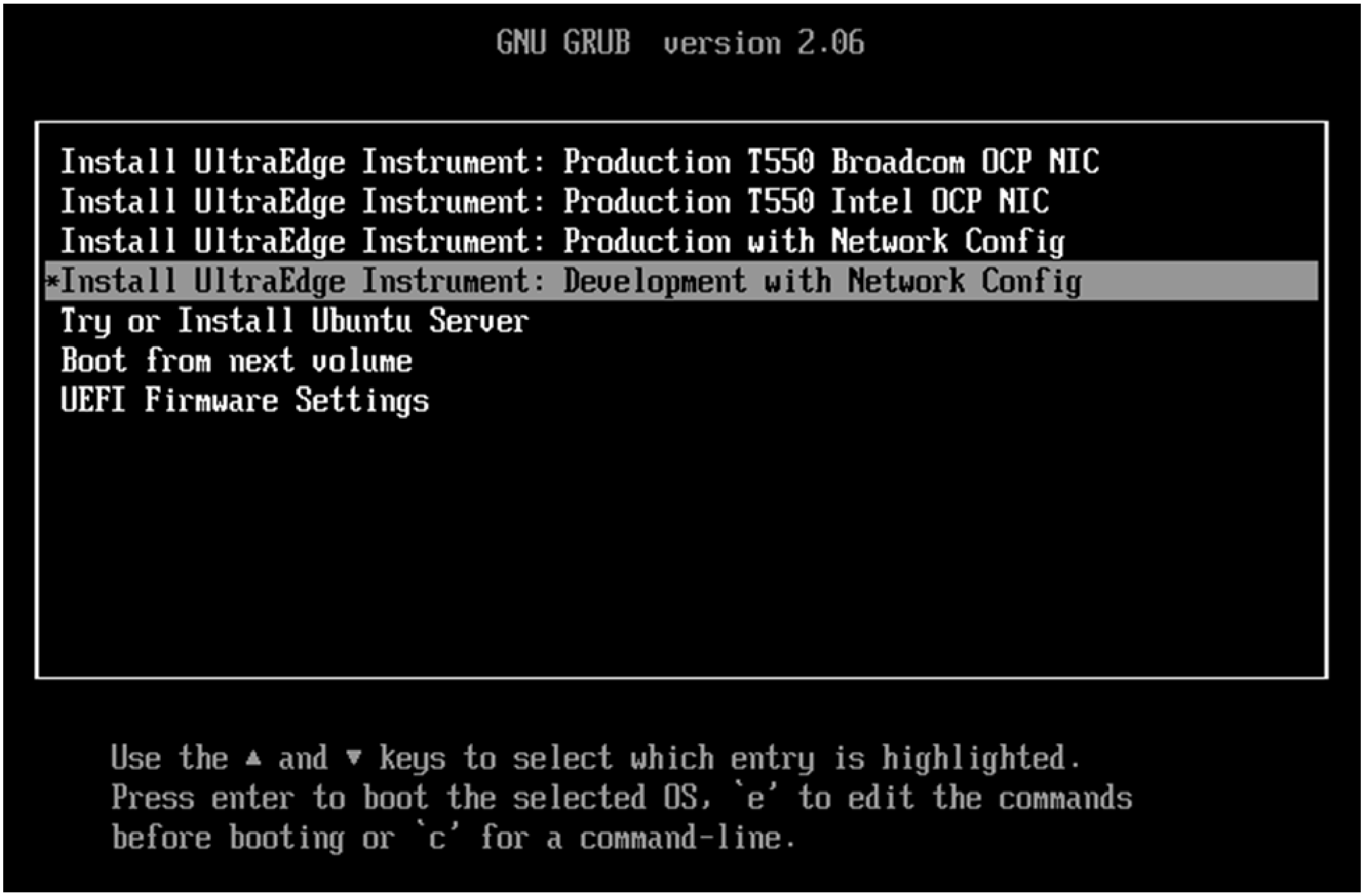
In the **Encryption Support** section, make sure to check the box entitled **Enable Trusted Platform Module.**





Linux Installation

Now that the Hyper-V virtual machine is created and configured, the UltraEdge Linux distribution can be installed. Turn on the virtual machine again. The correct boot options should be displayed. Select **Install UltraEdge Instrument: Development with Network Config**



Initial installation proceeds.

File Action Media Clipboard View Help

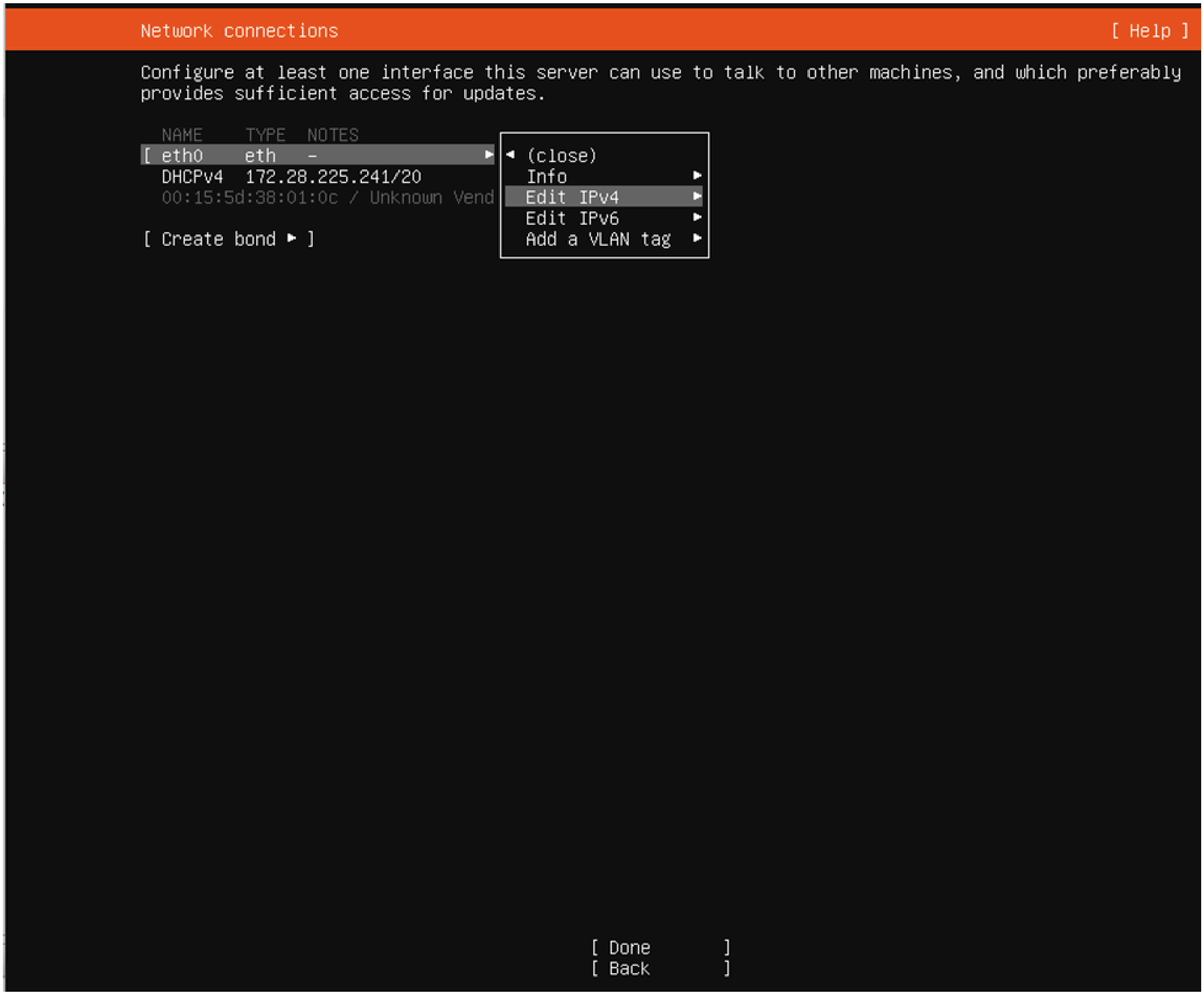
```
[  OK  ] Finished Load Kernel Module mtdpstore.
[    4.118452] alua: device handler registered
[  OK  ] Finished Load Kernel Module ramoops.
[    4.121877] emc: device handler registered
[    4.123504] rdac: device handler registered
[  OK  ] Finished Create Static Device Nodes in /dev.
[    4.128591] IPMI message handler: version 39.2
         Starting Rule-based Manager for Device Events and Files...
[    4.131752] ipmi device interface
[  OK  ] Finished Coldplug All udev Devices.
[  OK  ] Finished Load Kernel Modules.
         Starting Apply Kernel Variables...
[  OK  ] Finished Apply Kernel Variables.
[  OK  ] Started Rule-based Manager for Device Events and Files.
[  OK  ] Started Dispatch Password Requests to Console Directory Watch.
[  OK  ] Reached target Local Encrypted Volumes.
[  OK  ] Started Device-Mapper Multipath Device Controller.
[  OK  ] Reached target Preparation for Local File Systems.
         Mounting Mount unit for core20, revision 1587...
         Mounting Mount unit for lxd, revision 22923...
         Mounting Mount unit for snapd, revision 16292...
         Mounting Mount unit for subiquity, revision 3698...
         Mounting /tmp...
[  OK  ] Mounted /tmp.
[  OK  ] Mounted Mount unit for snapd, revision 16292.
[  OK  ] Mounted Mount unit for lxd, revision 22923.
[  OK  ] Mounted Mount unit for subiquity, revision 3698.
[  OK  ] Mounted Mount unit for core20, revision 1587.
[  OK  ] Reached target Local File Systems.
[  OK  ] Listening on Load/Save RF Kill Switch Status /dev/rfkill Watch.
         Starting Set console font and keymap...
         Starting Create final runtime dir for shutdown pivot root...
         Starting Tell Plymouth To Write Out Runtime Data...
         Starting Load AppArmor profiles managed internally by snapd...
         Starting Create Volatile Files and Directories...
         Starting Uncomplicated firewall...
[  OK  ] Finished Create final runtime dir for shutdown pivot root.
[  OK  ] Finished Tell Plymouth To Write Out Runtime Data.
[  OK  ] Finished Uncomplicated firewall.
[  OK  ] Finished Create Volatile Files and Directories.
         Starting Network Time Synchronization...
         Starting Record System Boot/Shutdown in UTMP...
         Starting Initial cloud-init job (pre-networking)...
[  OK  ] Finished Record System Boot/Shutdown in UTMP.
[  OK  ] Started Network Time Synchronization.
[  OK  ] Reached target System Time Set.
[  OK  ] Finished Set console font and keymap.
```

We now need to set the IP address of the UltraEdge VM.

Select **eth0** → **Edit IPv4**.

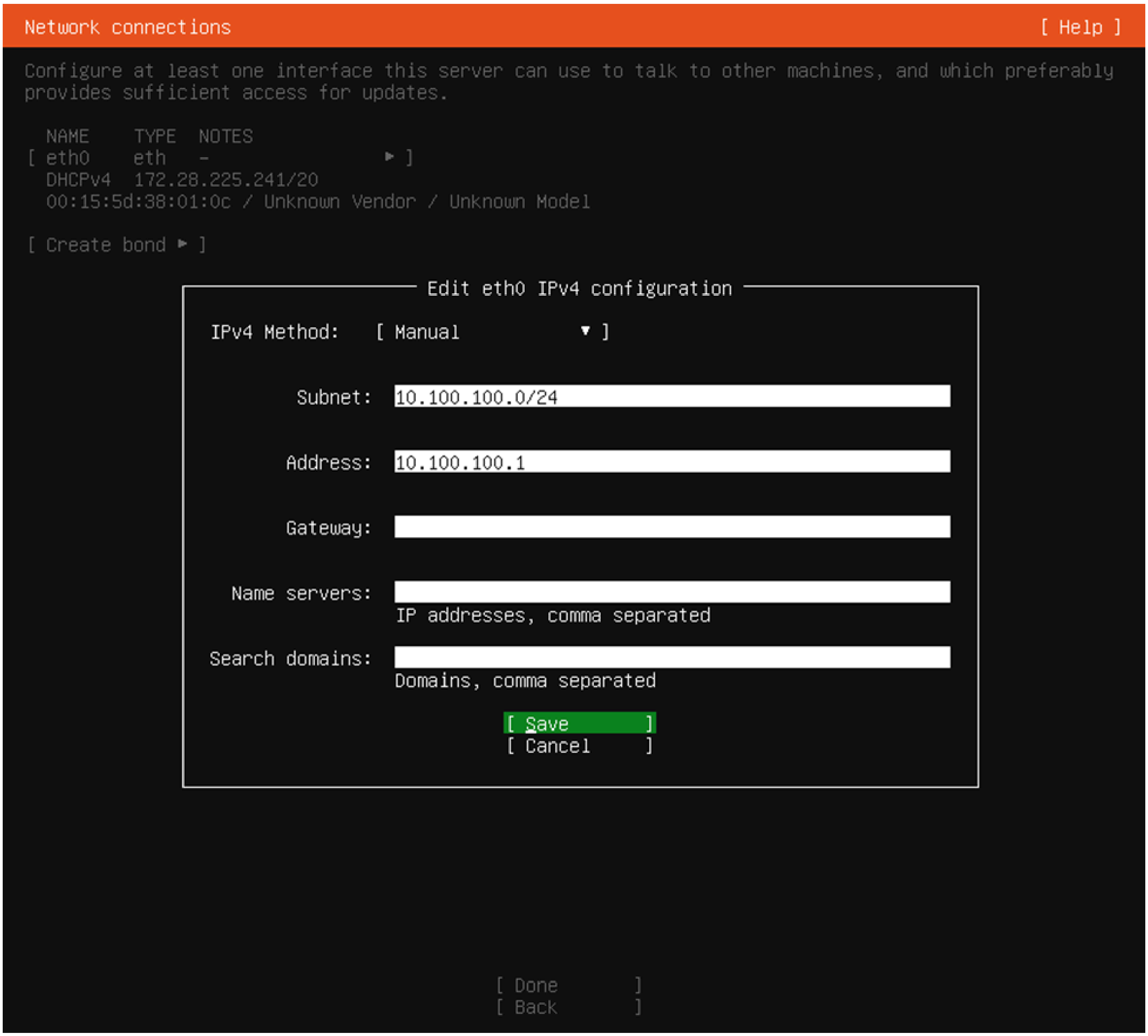
Note 1: Hit 'Enter' to select.

Note 2: The keypad is not available to type in numbers. Use the numbers above the letters on the keyboard.



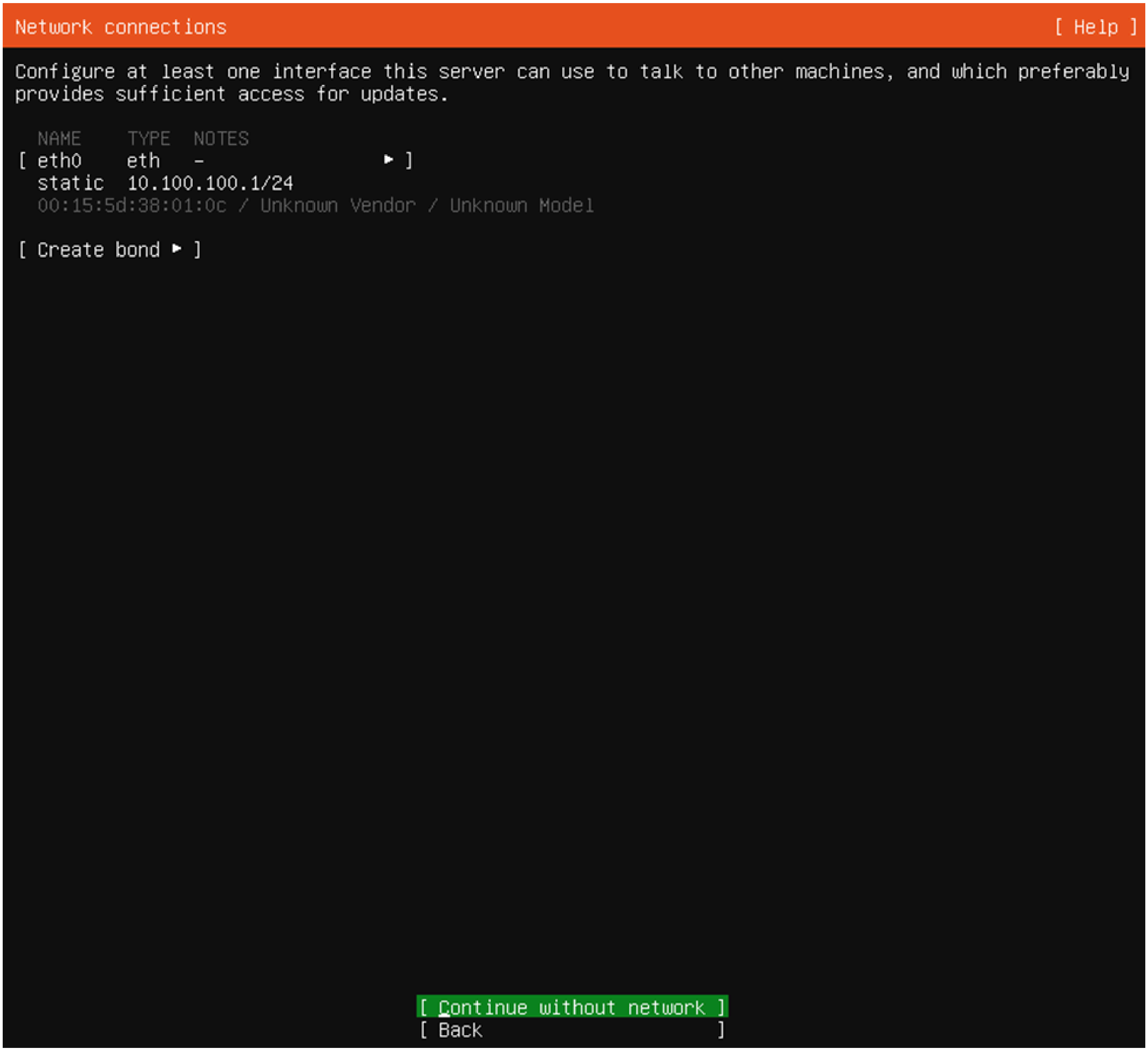
Then, select **Manual**

Fill the next screen as follows: Subnet must be **10.100.100.0/24** Address must be **10.100.100.1** Select **Save**

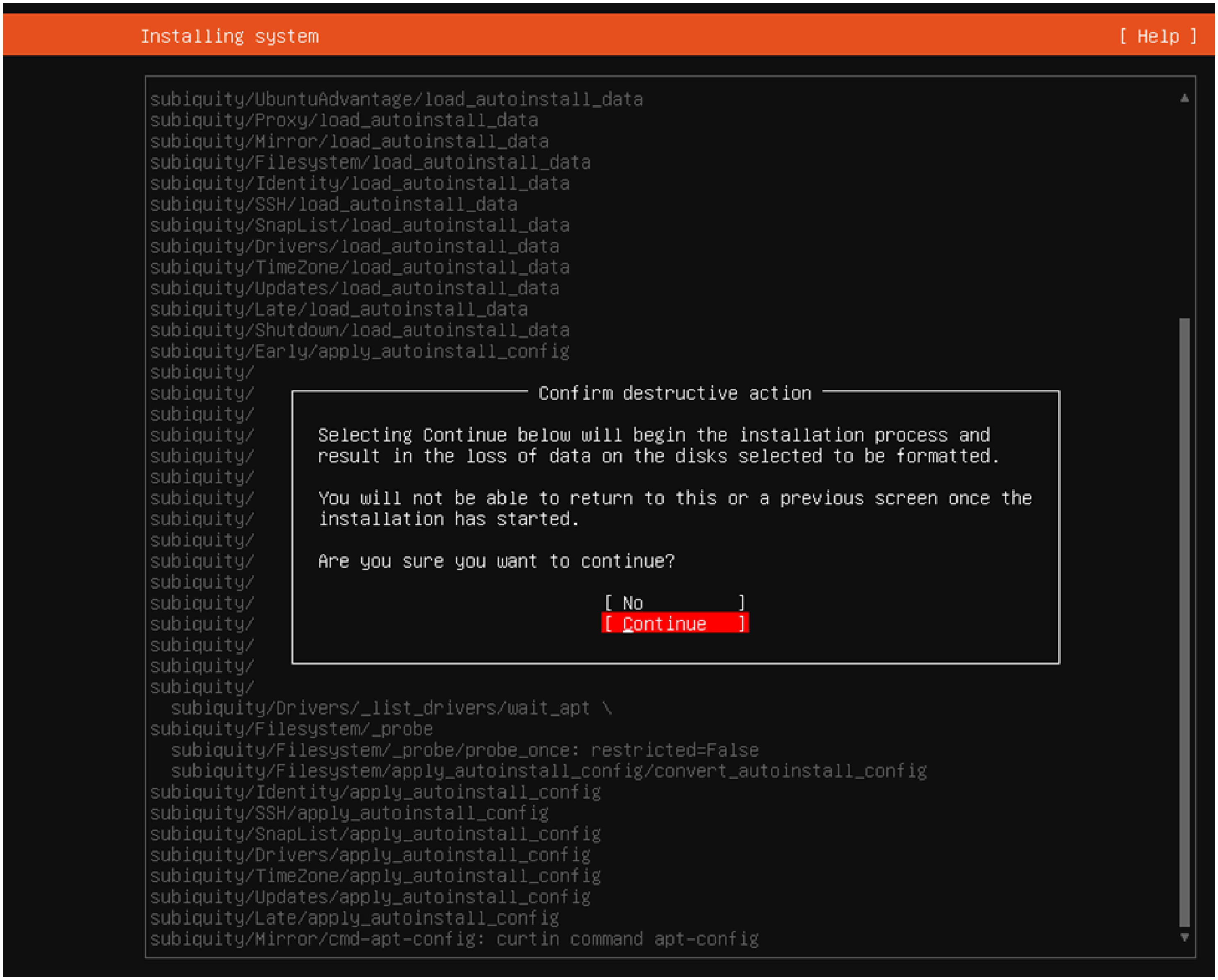


The new IP address is shown.

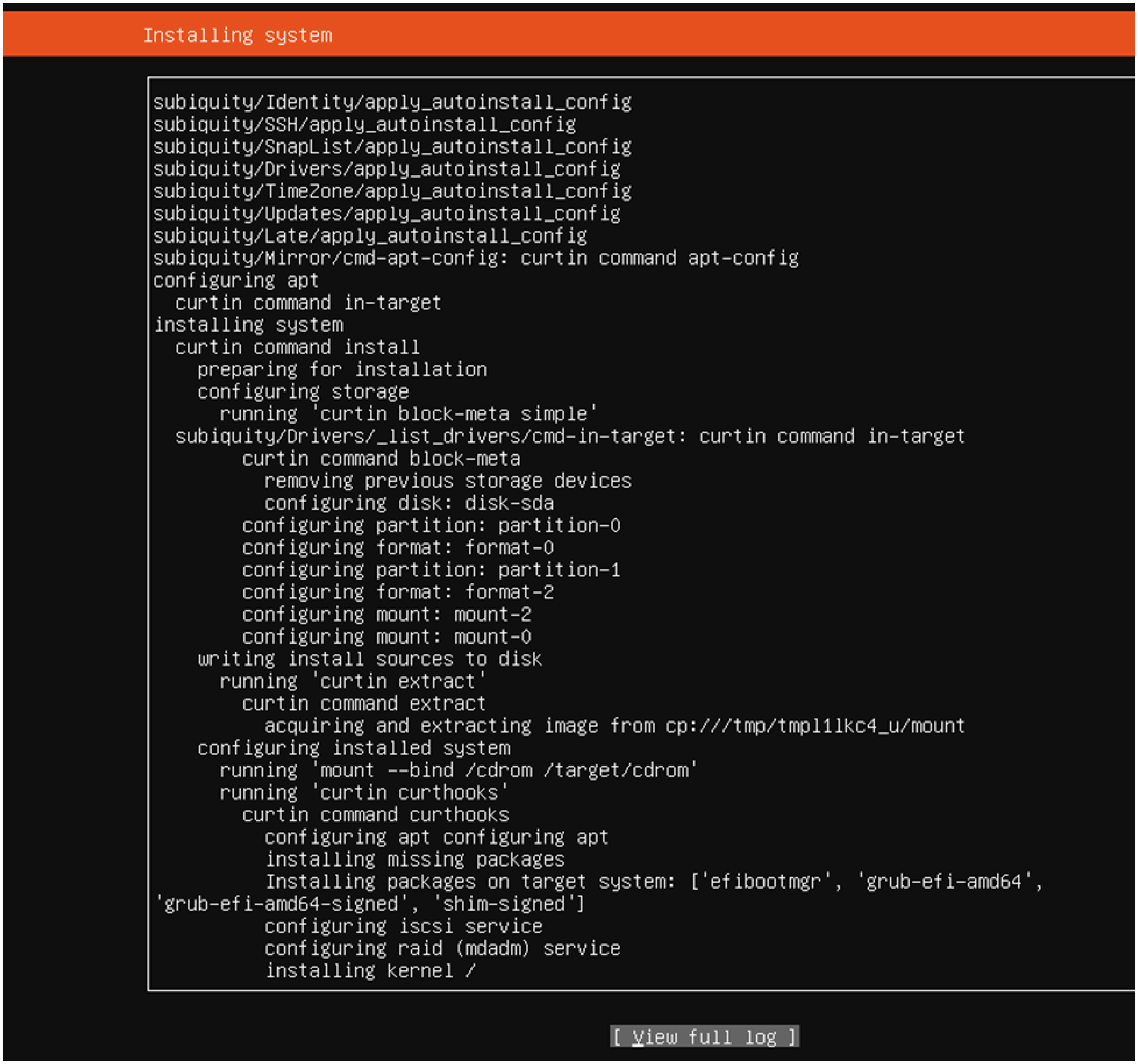
Select **Continue without network** (if the bottom of the screen is not visible, make sure your virtual machine is displayed 'Full screen').
Do not worry, this label is misleading. Network will be available.



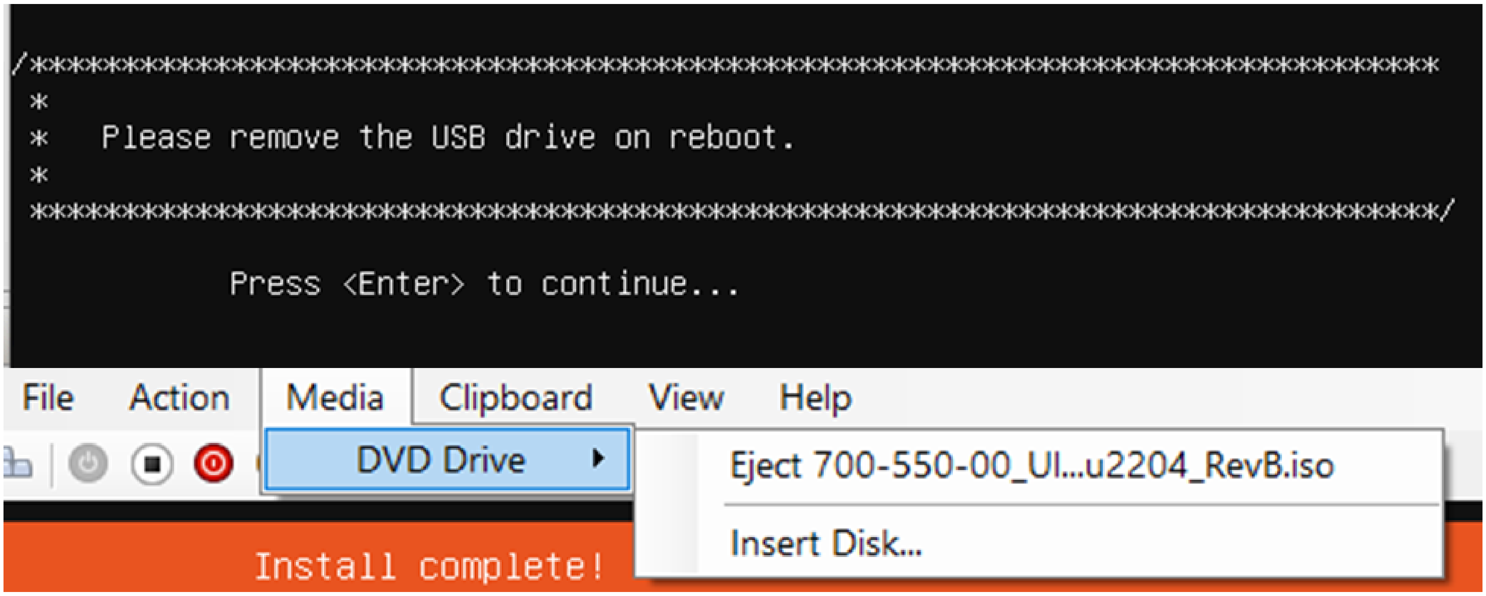
Confirm the action by choosing **Continue**



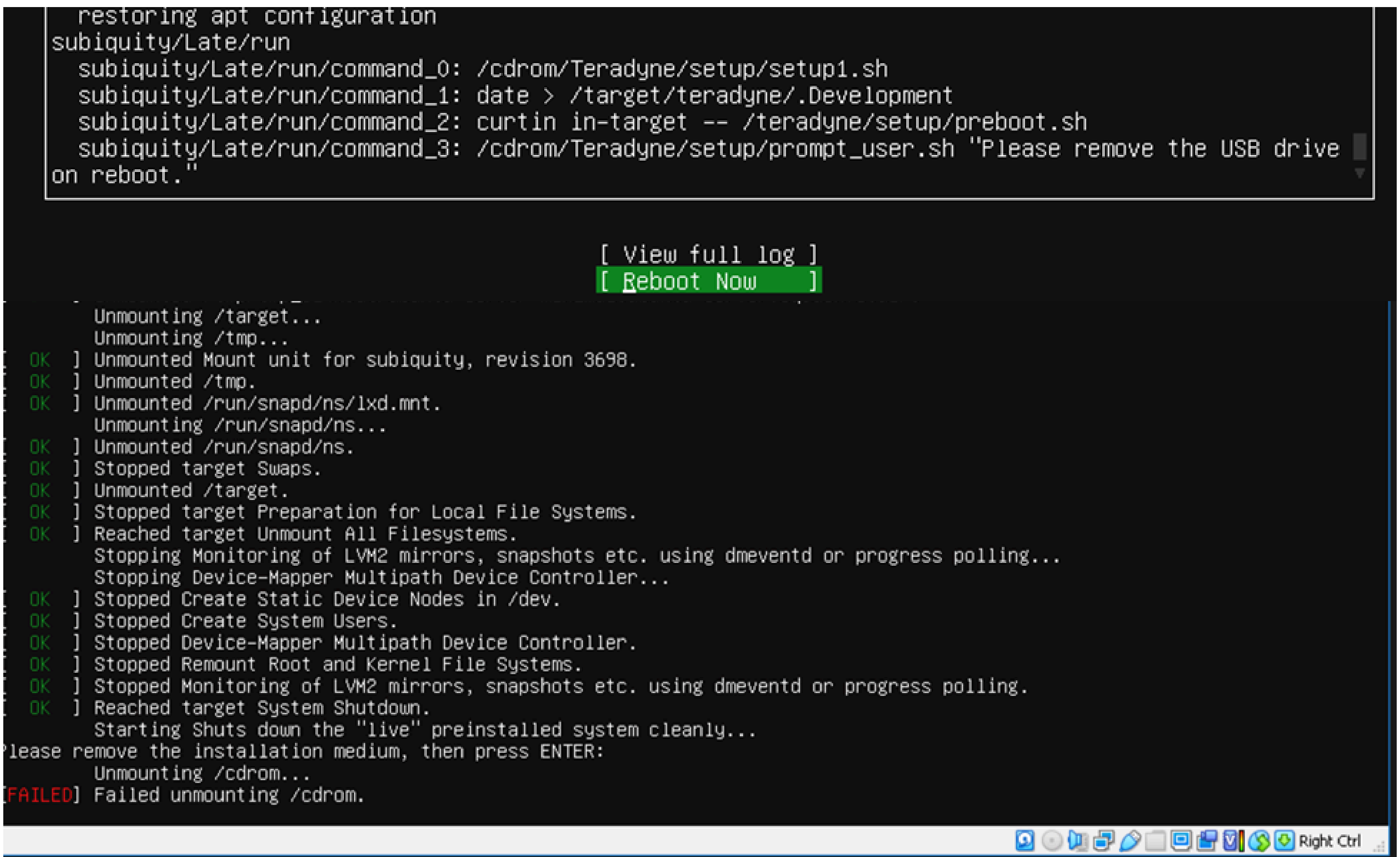
Installation continues until completion.



You might see a message asking to remove the drive before reboot. To do so, on the Virtual Machine Window, select **Media** → **DVD Drive** → **Eject ....** as shown here.



Select **Reboot Now** Errors may be displayed. These are linked to the mounting of the ISO file. Do not worry about them. Let the system run until setup completion.



```
se74jNb1n3wYn1ykOQRMviQ152QdgLqBoFck3uE= root@ultraedge
-----END SSH HOST KEY KEYS-----
[   51.736279] cloud-init[16436]: Cloud-init v. 22.4.2-0ubuntu0~22.04.1 finished at Thu, 28 Nov 2024 09:44:35 +0000. Data
DataSourceNone.  Up 51.73 seconds
[   51.739065] cloud-init[16436]: 2024-11-28 09:44:35,194 - cc_final_message.py[WARNING]: Used fallback datasource
[  OK  ] Finished Execute cloud user/final scripts.
[  OK  ] Reached target Cloud-init target.
[   52.271831] blk_update_request: I/O error, dev sr0, sector 2097136 op 0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[   52.272373] blk_update_request: I/O error, dev sr0, sector 2097136 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   52.272851] Buffer I/O error on dev sr0, logical block 2097136, async page read
[   52.273310] blk_update_request: I/O error, dev sr0, sector 2097137 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   52.273758] Buffer I/O error on dev sr0, logical block 2097137, async page read
[   52.274164] blk_update_request: I/O error, dev sr0, sector 2097138 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   52.274577] Buffer I/O error on dev sr0, logical block 2097138, async page read
[   52.274995] blk_update_request: I/O error, dev sr0, sector 2097139 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   52.275400] Buffer I/O error on dev sr0, logical block 2097139, async page read
[   52.275828] blk_update_request: I/O error, dev sr0, sector 2097140 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   52.276292] Buffer I/O error on dev sr0, logical block 2097140, async page read
[   52.276760] blk_update_request: I/O error, dev sr0, sector 2097141 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   52.277211] Buffer I/O error on dev sr0, logical block 2097141, async page read
[   52.277673] blk_update_request: I/O error, dev sr0, sector 2097142 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   52.278114] Buffer I/O error on dev sr0, logical block 2097142, async page read
[   52.278523] blk_update_request: I/O error, dev sr0, sector 2097143 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   52.279169] Buffer I/O error on dev sr0, logical block 2097143, async page read
[   52.279789] blk_update_request: I/O error, dev sr0, sector 0 op 0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[   52.280348] Buffer I/O error on dev sr0, logical block 0, async page read
[   52.280801] Buffer I/O error on dev sr0, logical block 1, async page read
[   59.304949] blk_update_request: I/O error, dev sr0, sector 2097136 op 0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[   59.305604] blk_update_request: I/O error, dev sr0, sector 2097136 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   59.306174] Buffer I/O error on dev sr0, logical block 2097136, async page read
[   59.306714] blk_update_request: I/O error, dev sr0, sector 2097137 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   59.307408] Buffer I/O error on dev sr0, logical block 2097137, async page read
[   59.307924] blk_update_request: I/O error, dev sr0, sector 2097138 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   59.308428] Buffer I/O error on dev sr0, logical block 2097138, async page read
[   59.308926] blk_update_request: I/O error, dev sr0, sector 2097139 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   59.309387] Buffer I/O error on dev sr0, logical block 2097139, async page read
[   59.309819] blk_update_request: I/O error, dev sr0, sector 2097140 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   59.310289] Buffer I/O error on dev sr0, logical block 2097140, async page read
[   59.310732] blk_update_request: I/O error, dev sr0, sector 2097141 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   59.311320] Buffer I/O error on dev sr0, logical block 2097141, async page read
[   59.311790] blk_update_request: I/O error, dev sr0, sector 2097142 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   59.312246] Buffer I/O error on dev sr0, logical block 2097142, async page read
[   59.312712] blk_update_request: I/O error, dev sr0, sector 2097143 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[   59.313162] Buffer I/O error on dev sr0, logical block 2097143, async page read
[   59.313686] blk_update_request: I/O error, dev sr0, sector 0 op 0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[   59.314182] Buffer I/O error on dev sr0, logical block 0, async page read
[   59.314704] Buffer I/O error on dev sr0, logical block 1, async page read
```

You are done

Enjoy